

VayuNetra — User Guide & Project Reference

The operations layer for urban air quality — trace who is polluting, predict what comes next, act with cited evidence, protect citizens in their own script. Ten Indian cities, ₹0 infrastructure.

Version of 18 August 2026 · Live app <https://vayunetra-aqi.vercel.app> · Source github.com/omkarr88/VayuNetra · Team DaGoats.

Every screenshot in this guide was captured by an automated walkthrough of the live console on 18 August 2026 (`web/scripts/qa/full-walkthrough.mjs`), so what you see here is what the application does — the same walkthrough checks every control and every download and passed with zero errors.

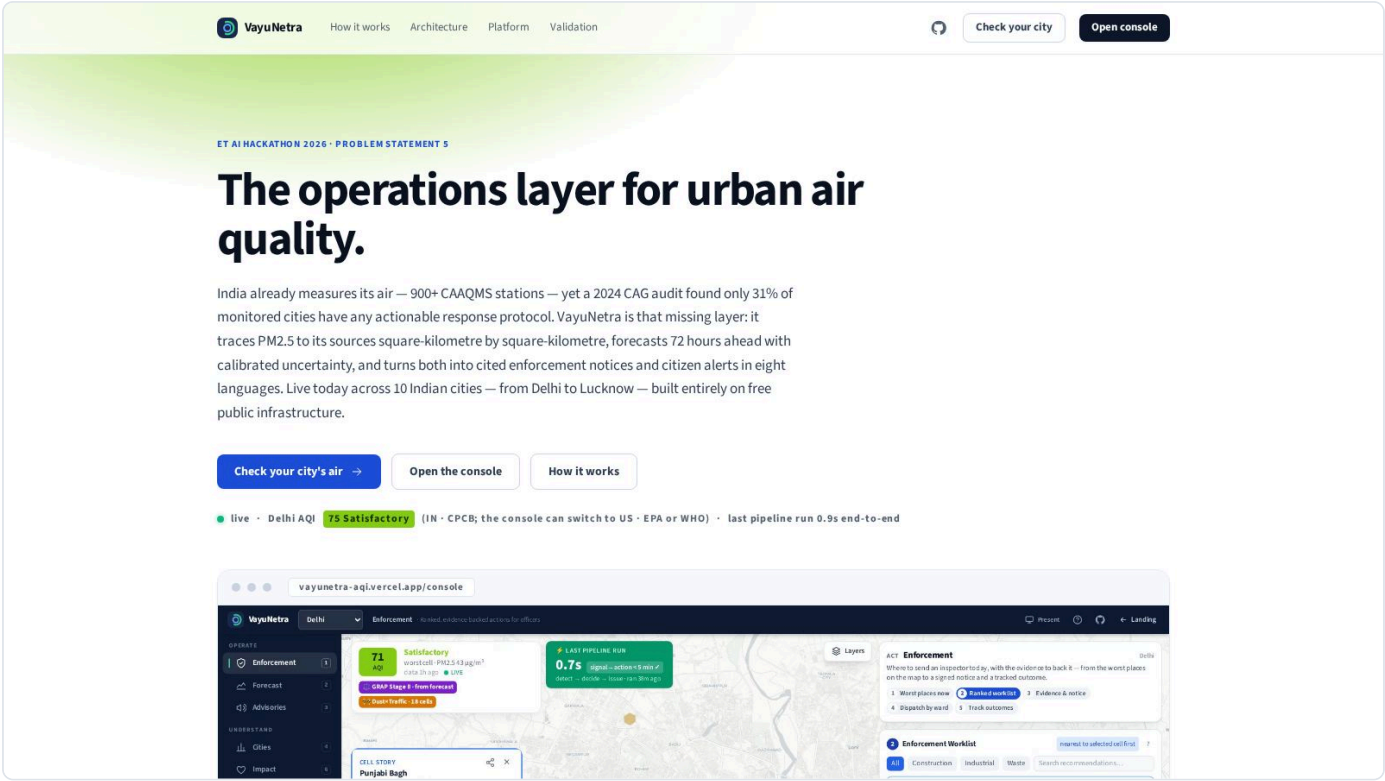
1. What VayuNetra is, in one minute

Indian cities already **measure** their air (900+ CAAQMS stations) and several **forecast** it. Nothing **acts** on it: a reading turns red and no system tells a city officer *who* is polluting this square kilometre right now, *what* the air will do tomorrow, or *where* to send inspectors first.

VayuNetra is that missing operations layer. Every ~1 km² cell in a city (an H3 resolution-8 hexagon) carries four things:

Verb	What the system does	Where you see it
Trace	Assigns PM2.5 blame per cell to traffic / construction dust / industrial / biomass burning / transported / other — a gradient-boosted model with SHAP explanations blended with chemical-signature priors, which abstains where it lacks out-of-sample skill	the map, the Cell Story
Predict	PM2.5 at +24 / 48 / 72 h with an 80 % conformal band and a calibrated probability of crossing Very Poor (>120 µg/m ³) or Severe (>250)	Cell Story tiles, Forecast section
Act	Ranks emission sources by contribution × people exposed × actionability × confidence; each with the real Sentinel-2 image of the site, the regulation retrieved verbatim, a draft notice PDF; the officer approves → dispatches → closes the case with a field finding on an immutable audit trail; every dispatch is measured before/after	Enforcement section
Protect	Templated (LLM-free) health advisories in eight scripts — Hindi, Kannada, Marathi, Tamil, Telugu, Bengali, Gujarati, English — over the app, a live Telegram bot, an IVR line and public displays	Advisories section

Six AI agents on one LangGraph do this in seconds of compute; the human steps (approval, dispatch, closure) are timestamped separately. It runs live in **Delhi, Bengaluru, Mumbai, Hyderabad, Chennai, Kolkata, Pune, Ahmedabad, Jaipur, Lucknow** on free tiers (Vercel + Render + Supabase + GitHub Actions). Adding a city is one YAML file.

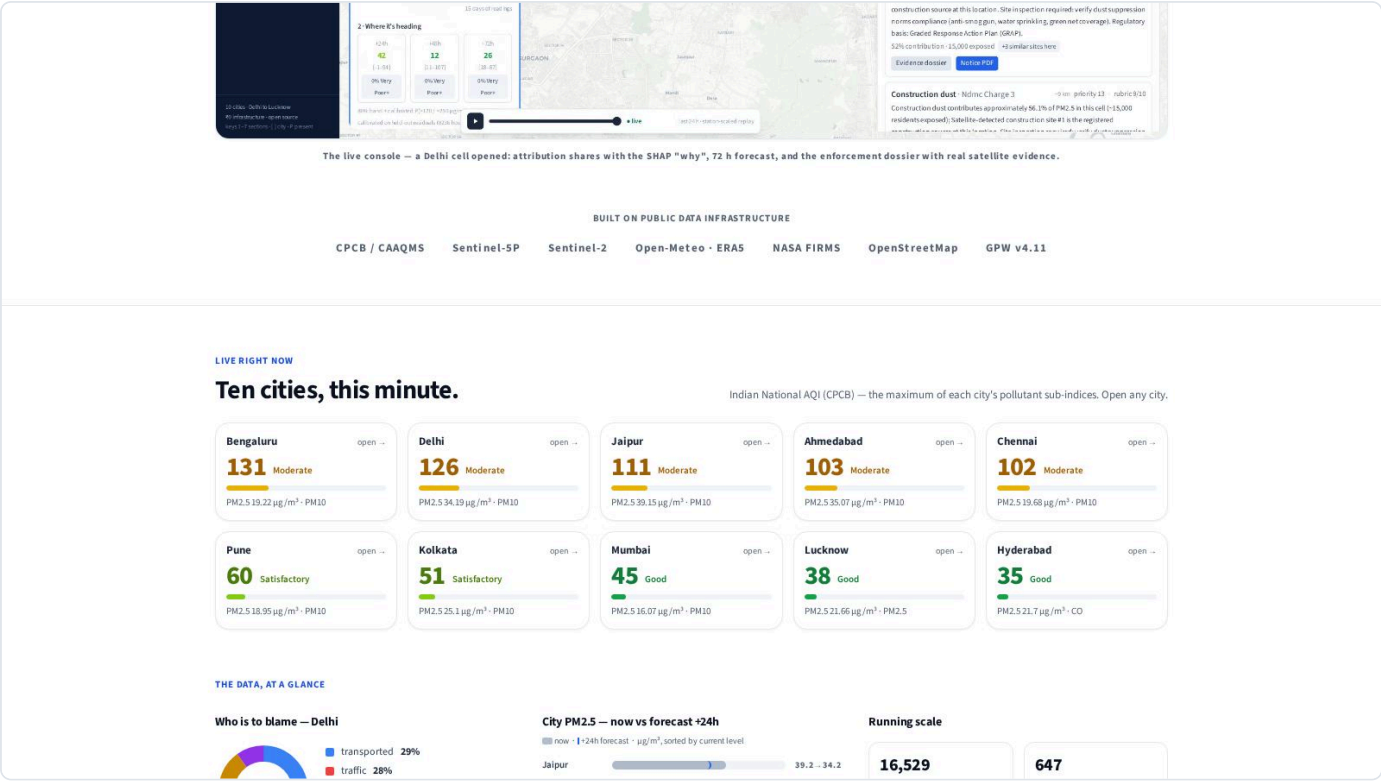


2. Getting in

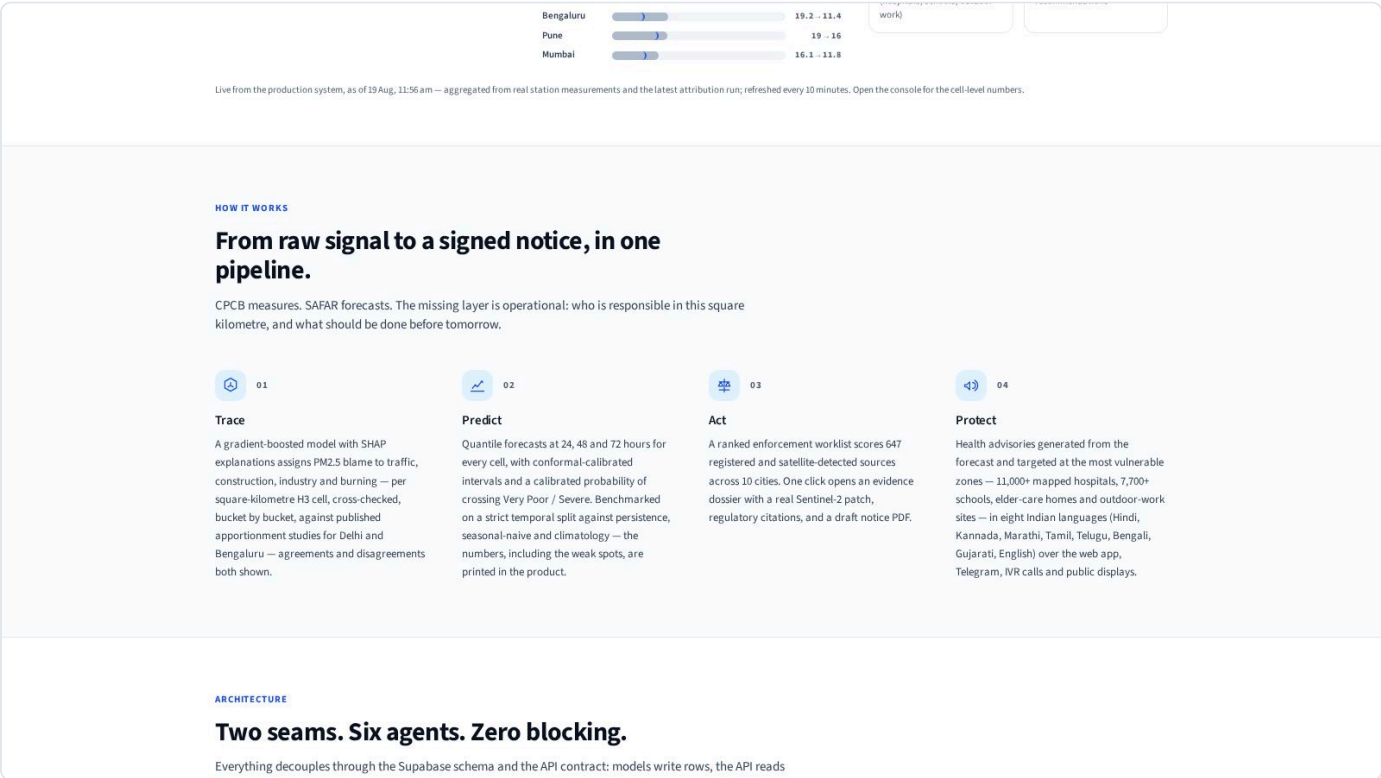
Front page	<code>https://vayunetra-aqi.vercel.app</code> — what VayuNetra is, with a live ten-city board. Two doors: Check your city's air (the public app) and Open console (operations).
Public app	<code>/city/<id></code> · <code>/map</code> · <code>/forecast</code> · <code>/rankings</code> · <code>/about</code> . Opens on Delhi. No login: reads are public under Postgres row-level security; every write goes through the server with the service role.
API	<code>https://vayunetra-c8i8.onrender.com</code> — <code>GET /health</code> shows <code>DEMO_MODE</code> ; every endpoint returns <code>{success, data, error, meta}</code> . Full list: <code>docs/API_CONTRACT.md</code> .
Run locally	<code>make dev</code> — starts the API on <code>:8000</code> (<code>DEMO_MODE</code> from <code>.env</code>) and the web app on <code>:5173</code> . <code>make prewarm</code> wakes and verifies the live stack (GO / NO-GO). <code>make test</code> runs the 204 backend tests.
Demo mode	<code>DEMO_MODE=true</code> serves 19 bundled fixtures with an identical UX (labelled in <code>/health</code>). The web app also falls back to bundled fixtures if the API is unreachable and shows an amber <i>"backend waking up — showing bundled demo snapshot"</i> notice (Render free tier sleeps; retry or wait — it clears on the first successful call).
Keyboard	In the console: 1–7 switch sections · [] cycle cities · P presentation mode (larger type for a projector) · ⌘K / Ctrl-K jump to any city or section · Esc closes the Cell Story.
Deep links	Every console state is a URL: <code>/console?city=mumbai&section=forecast&cell=88608b56cbffff&mode=pm25&layers=sources,plumes,wards,freight,fires</code> .

3. The landing page (/)

The public front door — for citizens, judges and officers who have not opened the console yet. It reads live data where it can.



- **Hero** — the one-sentence proposition, **Open the console**, and the console screenshot.
- **The data, at a glance** — read live from `GET /landing/snapshot` (10-minute cache, stamped *as of HH:MM*): the Delhi source-mix donut (mean of the latest attribution run), every city's PM2.5 now vs +24 h forecast (sorted worst first), and the running scale — ~1 km² cells modelled, registered + satellite-detected sources, vulnerability-scored zones, live enforcement recommendations. A bundled snapshot is shown only until the API answers.



- **How it works** — trace → predict → act → protect, with the six agents and the data sources.
- **Architecture** — the two-seam architecture diagram (schema seam, API seam) — the same SVG the pitch deck uses.
- **Platform** — what runs where (Vercel, Render, Supabase, GitHub Actions) and the ₹0 point.

- **Validation** — the claim / result / how-it-was-checked table: attribution vs published apportionment (cosine 0.88 / 0.90 / 0.93 plus bucket tables), forecast skill vs persistence, the calibrated probabilities, the rejected TFT model, test counts.

VALIDATION

Every number is checked — including the failures.

Each claim below is reproducible from the repository — the evaluation notebook and `python -m ml_eval.benchmark`, whose artifacts the API serves and the console prints. Where a method underperformed, that result ships too.

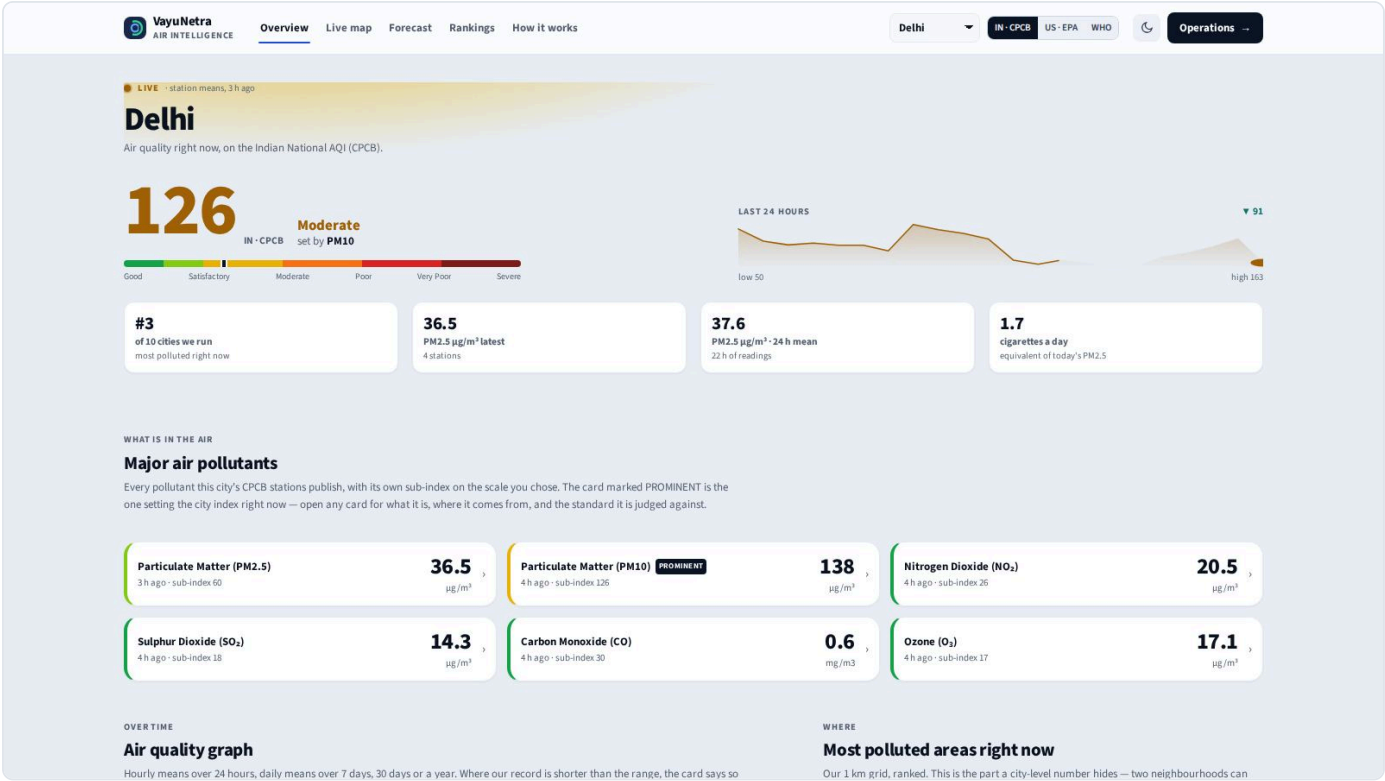
CLAIM	RESULT	METHOD
Attribution checked against published apportionment	cosine 0.88 / 0.90 / 0.93	vs SAFAR-Delhi 2018, CSTEP-Bengaluru 2022 (verified from the report), Urban-Emissions Mumbai — plus bucket-by-bucket tables vs TERI-ARAI and Guttikunda et al.
Attribution behaves physically	2.30x traffic signal in rush hours	IST rush vs off-peak SHAP; weather controlled
Forecast beats real baselines	Delhi +9 / +13 / +12% · Mumbai +17 / +19 / +21% at 24/48/72 h	multi-season temporal split (2025-26 winter + summer 2026), monthly refit on the trailing 90 d; served forecast = LightGBM blended with persistence; vs persistence and weekly seasonal-naïve — the raw model's weaker numbers ship too
It warns before the air turns	51-54% of Very-Poor onsets flagged 1-3 days ahead	alarm on the calibrated probability $P \geq 0.3$ (precision 0.64-0.68); persistence catches 0% by construction; the Severe tail stays weak — stated, not hidden
Uncertainty is calibrated	80% band — 78% measured · $P(>120)$ Brier skill +51%	conformal calibration; exceedance probabilities on every cell forecast drive the alarms
Enforcement is equitable	no socio-economic inputs, by construction	fairness audit on every live recommendation (n=390 at the July audit)
Model choice was earned	TFT trained on GPU — and rejected	LightGBM won every launch city on held-out skill
The loop is fast	seconds from signal to cited recommendation	live per-node agent traces; approval, dispatch and closure timestamped per action

4. One product, three surfaces

VayuNetra is one application wearing three faces, and they share a design system and a navigation bar so that moving between them never moves the furniture:

Surface	Path	Who it is for
Front page	/	Anyone arriving cold — what this is, and a live ten-city board
Public app	/city/<id> , /map , /forecast , /rankings , /about	A resident asking "what is the air here, and what should I do"
Operations console	/console?section=...	An officer deciding where to send an inspector today

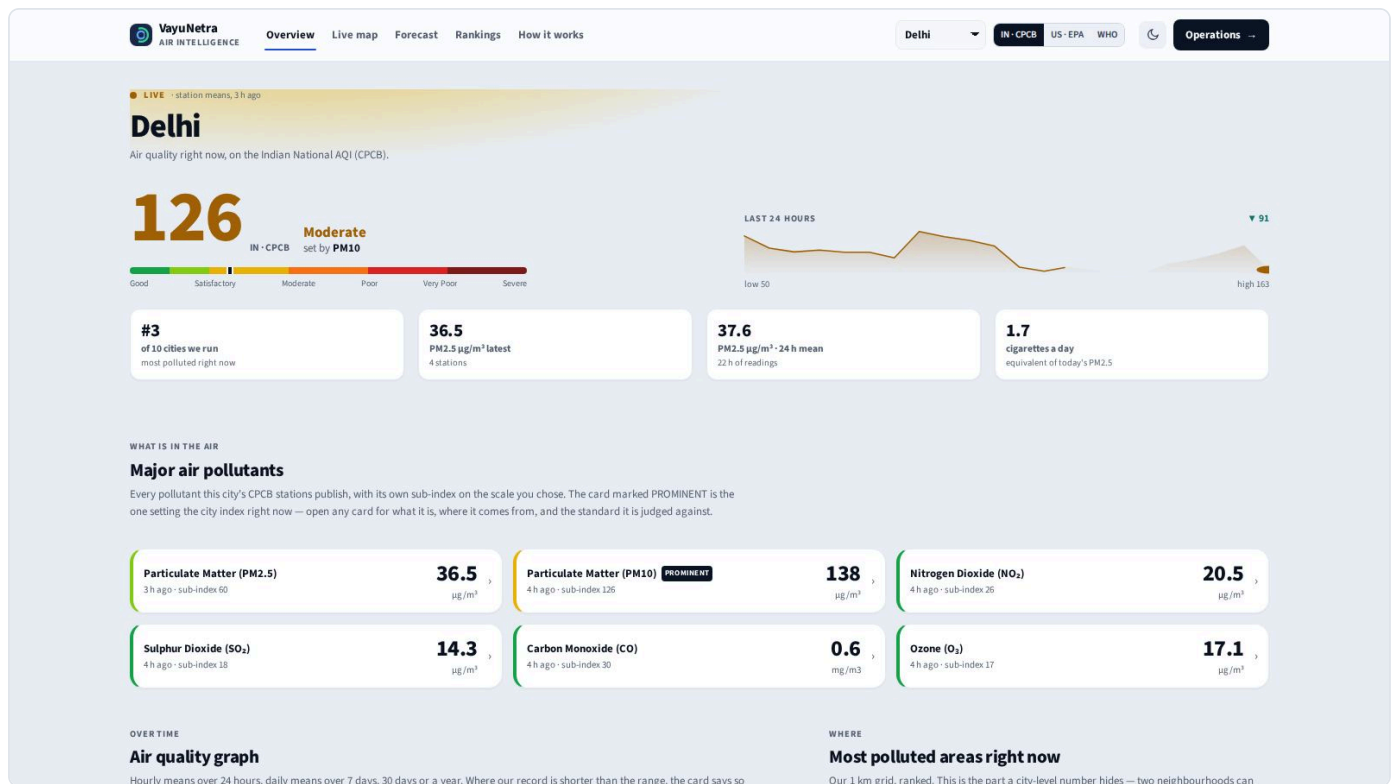
The same bar sits at the top of every page: the VayuNetra mark (back to the front page), the section links, the **city selector**, the **scale toggle** (IN · CPCB / US · EPA / WHO), the light/dark switch, and one button that crosses between the public app and the console — **Operations** → on public pages, **Public site** → in the console.



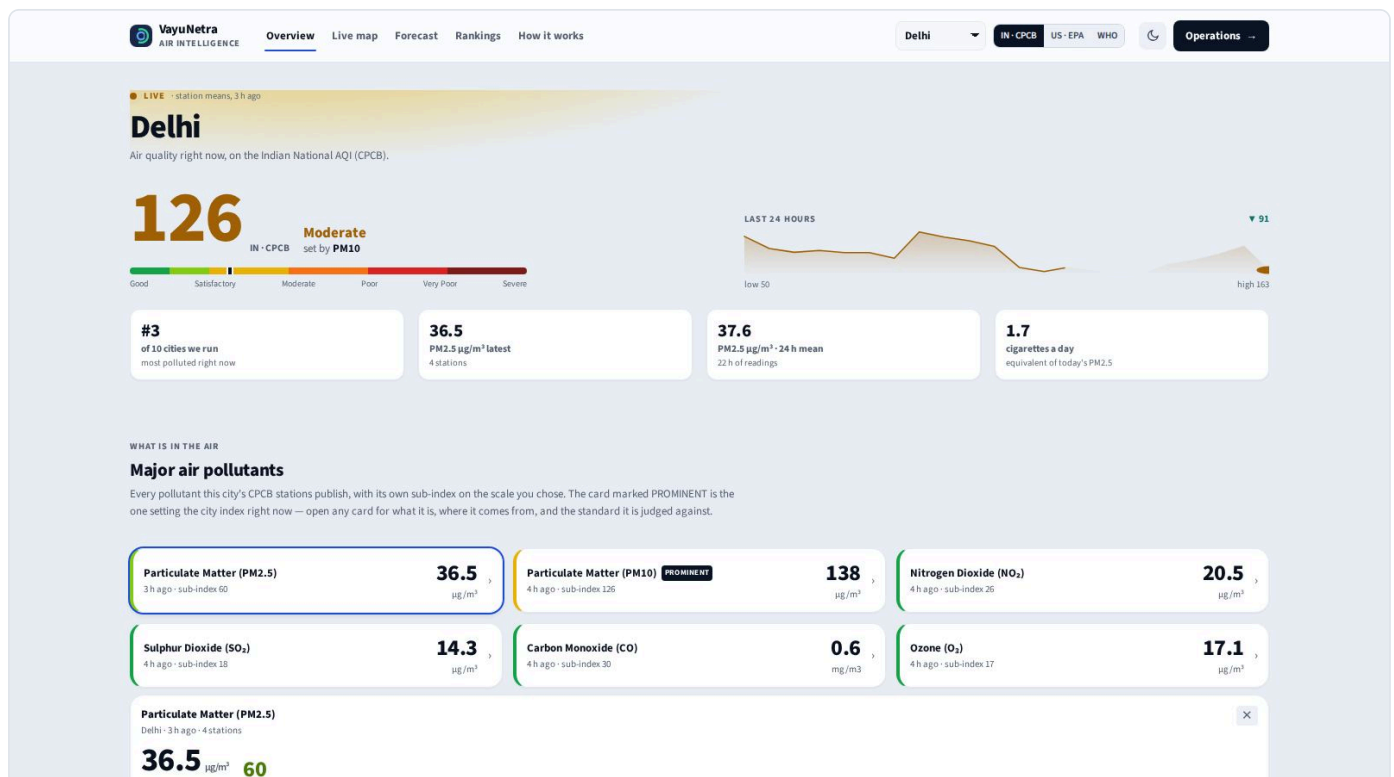
4.1 The public app

Five pages, all following the chosen city:

Page	What it answers
Overview (/city/<id>)	The city's index right now, every pollutant, the graph, where it is worst, the calendar, the yearly rhythm, health advice, and the other cities. No map, by design — this page is for reading numbers.
Live map (/map)	The 1 km grid, full-bleed, with the layer control and the cell story.
Forecast (/forecast)	The 72-hour outlook and the city's source mix — what happens next and why.
Rankings (/rankings)	All ten cities live, sortable, plus the three-scale explainer.
How it works (/about)	The five-step pipeline, the scales, every data source, and what the system is <i>not</i> .

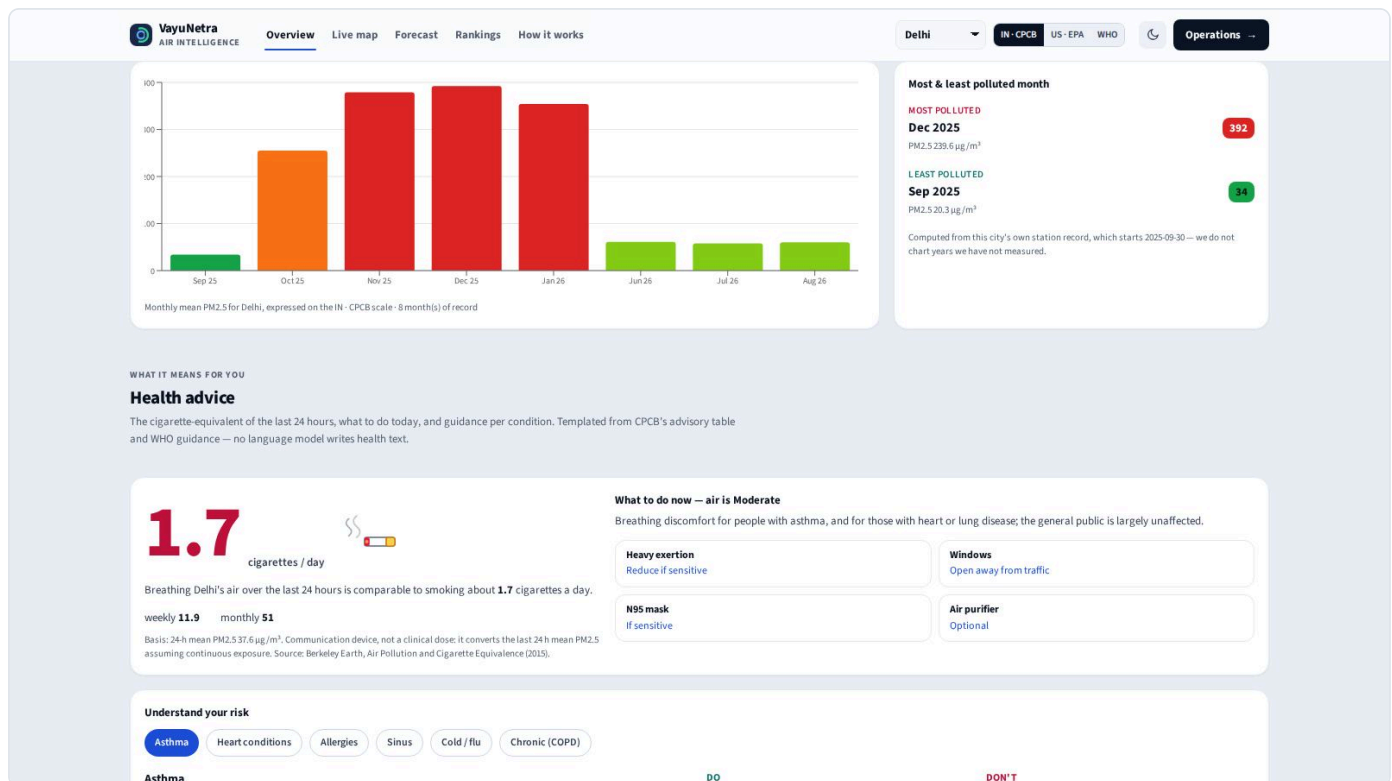


Each pollutant card carries its own sub-index on the chosen scale; the one marked **PROMINENT** is setting the city index right now. Opening a card gives what the pollutant is, where it comes from in an Indian city, what it does to people, and the CPCB and WHO standards it is judged against.

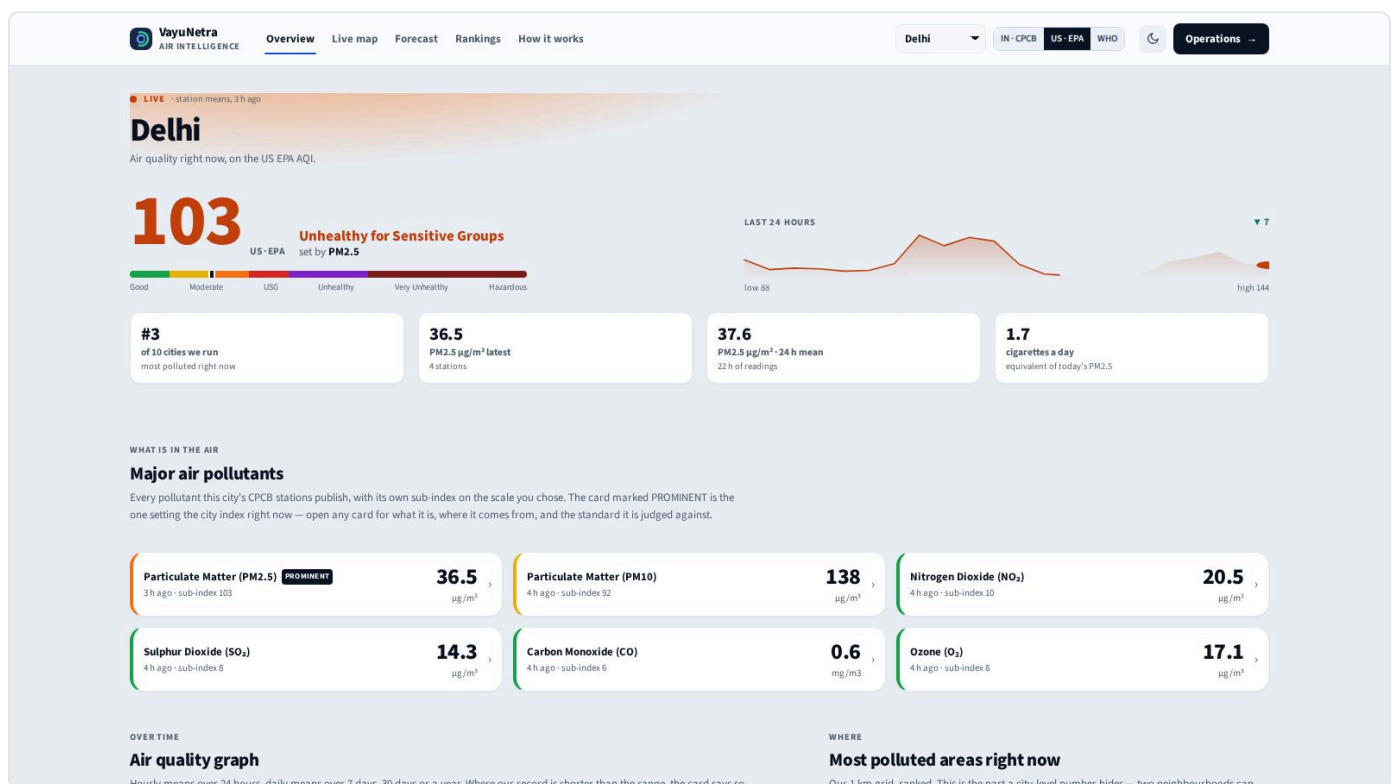


The graph runs 24 h (hourly means) or 7 / 30 / 365 days (daily means), for the index or any single pollutant. Beside it, **most polluted areas right now** ranks the city's own 1 km cells — the part a city-level number hides.

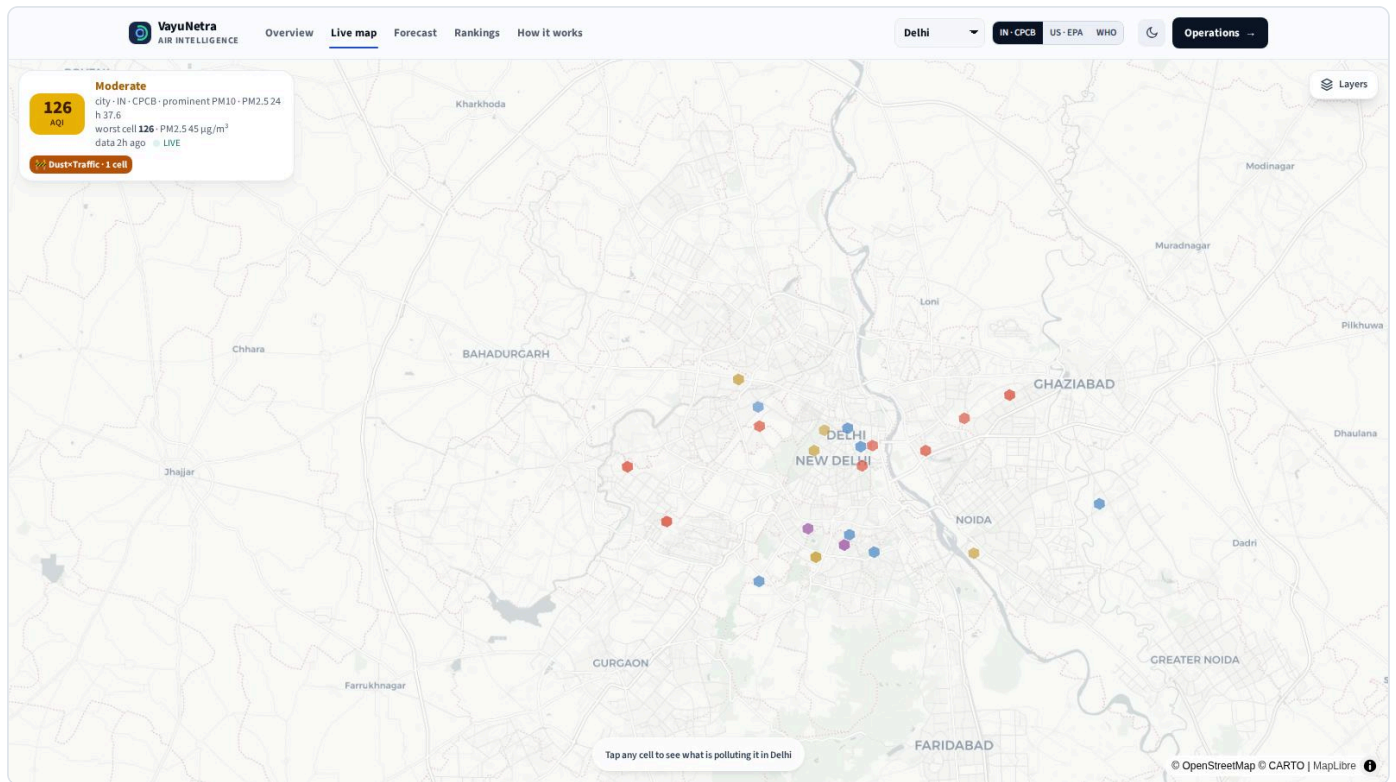
THE YEAR



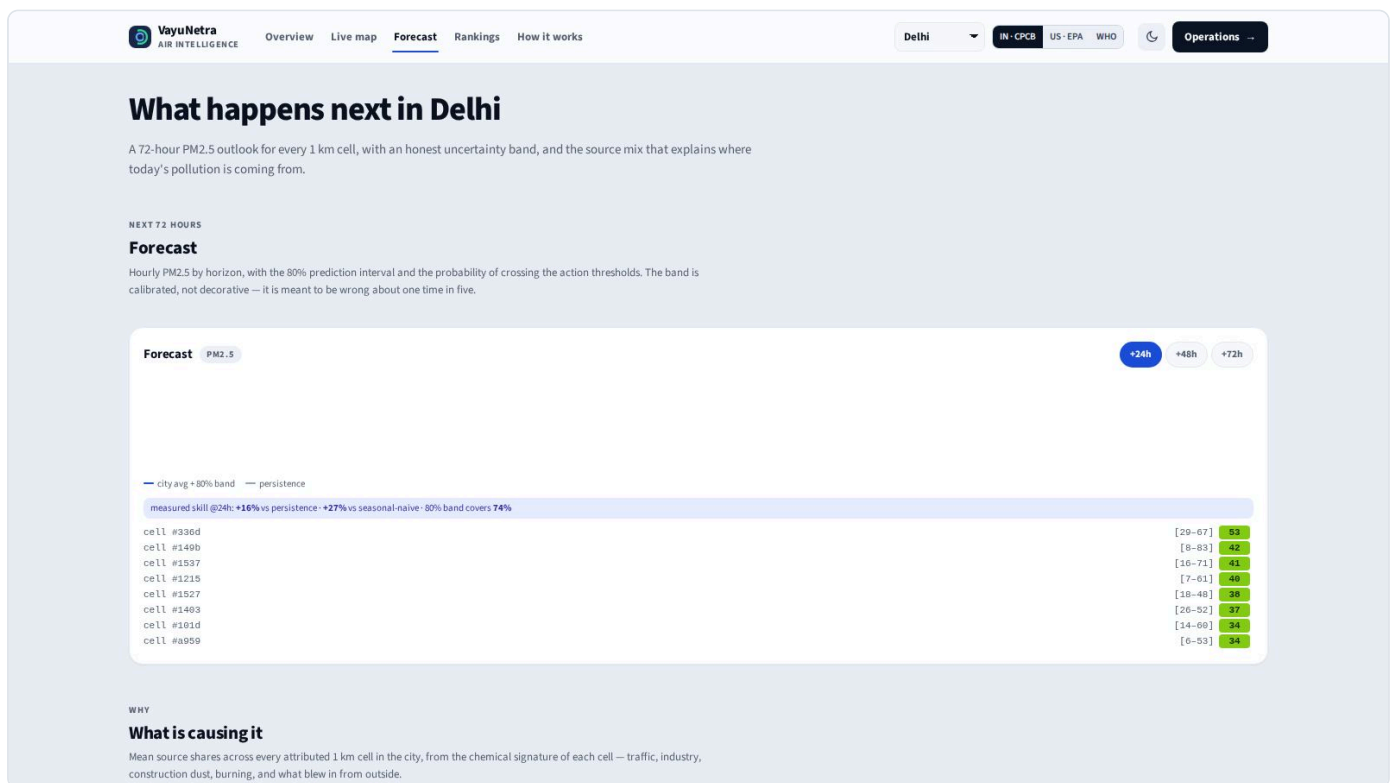
Switching the scale changes the labels, the band colours and the index number — never the measured concentration underneath. The same air, read on the US EPA scale:



The **Live map** is the same MapLibre + deck.gl view the console uses, full-bleed, with the layer control and the cell story:



Forecast pairs the 72-hour outlook with the city's mean source mix — what happens next, and why:



Rankings puts all ten cities on one board, and explains the three scales side by side:

VayuNetra
AIR INTELLIGENCE

OverviewLive mapForecastRankingsHow it works

DelhiIN - CPCBUS - EPAWHOOperations

India, right now

Ten cities, one definition of the index, updated every hour from CPCB station data. Sort by any column; click a city to open its page.

ON THE INDIAN NATIONAL AQI (CPCB)

Live city rankings

Kolkata: construction dust · Bengaluru: industrial · Mumbai: construction dust · Delhi: transported · Pune: construction dust · Lucknow: industrial · Chennai: construction dust · Ahmedabad: transported · Hyderabad: industrial · Jaipur: transported

city	status	IN - CPCB	PM2.5 $\mu\text{g}/\text{m}^3$	+24 h PM2.5	dominant source
Bengaluru	Moderate	131 PM10	19.22	11.44	industrial
Delhi	Moderate	126 PM10	34.19	32.23	transported
Jaipur	Moderate	111 PM10	39.15	34.19	transported
Ahmedabad	Moderate	103 PM10	35.07	25.55	transported
Chennai	Moderate	102 PM10	19.68	21.48	construction dust
Pune	Satisfactory	60 PM10	18.95	16.04	construction dust
Kolkata	Satisfactory	51 PM10	25.1	22.53	construction dust
Mumbai	Good	45 PM10	16.07	11.78	construction dust
Lucknow	Good	38 PM2.5	21.66	18.97	industrial
Hyderabad	Good	35 CO	21.7	18.6	industrial

Kolkata51

Bengaluru131

Mumbai45

Delhi126

VayuNetra
AIR INTELLIGENCE

OverviewLive mapForecastRankingsHow it works

DelhiIN - CPCBUS - EPAWHOOperations

Pune60SatisfactoryPM2.5 18.95+24 h 16.04

Lucknow38GoodPM2.5 21.66+24 h 18.97

Chennai102ModeratePM2.5 19.68+24 h 21.48

Ahmedabad103ModeratePM2.5 35.07+24 h 25.55

Hyderabad35GoodPM2.5 21.7+24 h 18.6

Jaipur111ModeratePM2.5 39.15+24 h 34.19

A city appears only once its stations have reported in the current window. *+24 h* is the forecast city-mean PM2.5, not an index.

READ THE NUMBER

Which scale am I looking at?

The same air, three published yardsticks. Switching the toggle in the header changes labels, band colours and the index number — never the measured concentration underneath.

IN - CPCB

Indian National AQI (CPCB)

SHOWING

Each pollutant gets a sub-index from its own breakpoints; the city index is the highest of them, and that pollutant is named the prominent one.

Bands. Good · Satisfactory · Moderate · Poor · Very Poor · Severe

This is the number Indian officers, CPCB bulletins and GRAP actions run on. It is our default for that reason.

US - EPA

US EPA AQI

switch to this

Same idea, different breakpoints — the 2024 PM2.5 revision makes it stricter at the clean end, so the US number usually reads higher than the Indian one for the same air.

Bands. Good · Moderate · Unhealthy for Sensitive Groups · Unhealthy · Very Unhealthy · Hazardous

Most international apps display this, so it is here to let you check us against them without changing what we measure.

WHO

WHO 2021 guideline

switch to this

Not an index at all — PM2.5 expressed as a multiple of the WHO 24-hour guideline of 15 $\mu\text{g}/\text{m}^3$, banded by the WHO interim targets.

Bands. Within guideline · above it, by interim target IT-4 down to IT-1

The health yardstick: it says how far from safe the air is, rather than which administrative box it falls in.

VayuNetraPAGESFOR OFFICIALS

How it works states the pipeline, names every data source, and — as importantly — says what the system is not: not a compliance measurement, not a legal finding, not medical advice.

[Overview](#)
[Live map](#)
[Forecast](#)
[Rankings](#)
[How it works](#)

Delhi

IN · CPCB

US · EPA

WHO

Operations

Three scales, one measurement

The toggle in the header changes how the number is expressed, never what was measured.

IN · CPCB

SHOWING

Indian National AQI (CPCB)

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The health yardstick; it says how far from safe the air is, rather than which administrative box it falls in.

PROVENANCE

Where the data comes from

Every source is public. None of it is synthesised, and nothing on this site is filled in when a station is silent — gaps are shown as gaps.

CPCB / CAAQMS	India's official continuous ambient air-quality monitoring stations — the ground truth for every index on this site.
OpenAQ	Aggregated public station data, used to widen coverage where CAAQMS is thin.
Copernicus Sentinel-5P	Tropospheric NO ₂ column density — the satellite view of combustion, used as an attribution feature.
NASA FIRMS (MODIS / VIIRS)	Active-fire detections — the burning signal behind the fire layer on the map.
Open-Meteo	Weather and boundary-layer variables that drive both the forecast and the deweathering.
OpenStreetMap · WorldPop	Roads, land use, industrial sites and population — the static layers behind exposure and enforcement targeting.

VayuNetra OPERATIONS
Enforcement
Forecast
Advisories
Cities
Simulator
Impact
Pipeline
Delhi ▾
IN · CPCB
US · EPA
WHO
Present
Tour
🌙
Public site ⋮

ACT - DELHI

Enforcement

Where to send an inspector today, with the evidence to back it — from the worst places on the map to a signed note about the outcome.

1 Morning brief
2 Ranked workload
3 Evidence & notice
4 Dispatch by ward
5 Track outcomes

Moderate

city: IN · CPCB · prominent PM₁₀ · PM_{2.5} 24 h 37.6

CELL STORY

Malviya Nagar

~1 km² cell · 863da11527ffff

1· Who's to blame — Construction Dust conf 77%

Source	Percentage
Construction Dust	77%
Traffic	12%
Transported	7%
Other	9%
Industrial	6%
Biomass Burning	6%

Why — chemical-signature attribution

Marker chemistry points to construction dust and traffic.

Local model missed the 30_15 skill gate here — we fall back to cited chemical-signature priors rather than over-claim.

evidence: NO₂ 10 · PM₁₀/PM_{2.5} 4.5

THIS PLACE — PAST AIR

30d 90d 1y

... About the same as a month ago (+2%) · mostly satisfactory over the last 30 days

Layers

© OpenStreetMap © CARTO | MapLibre

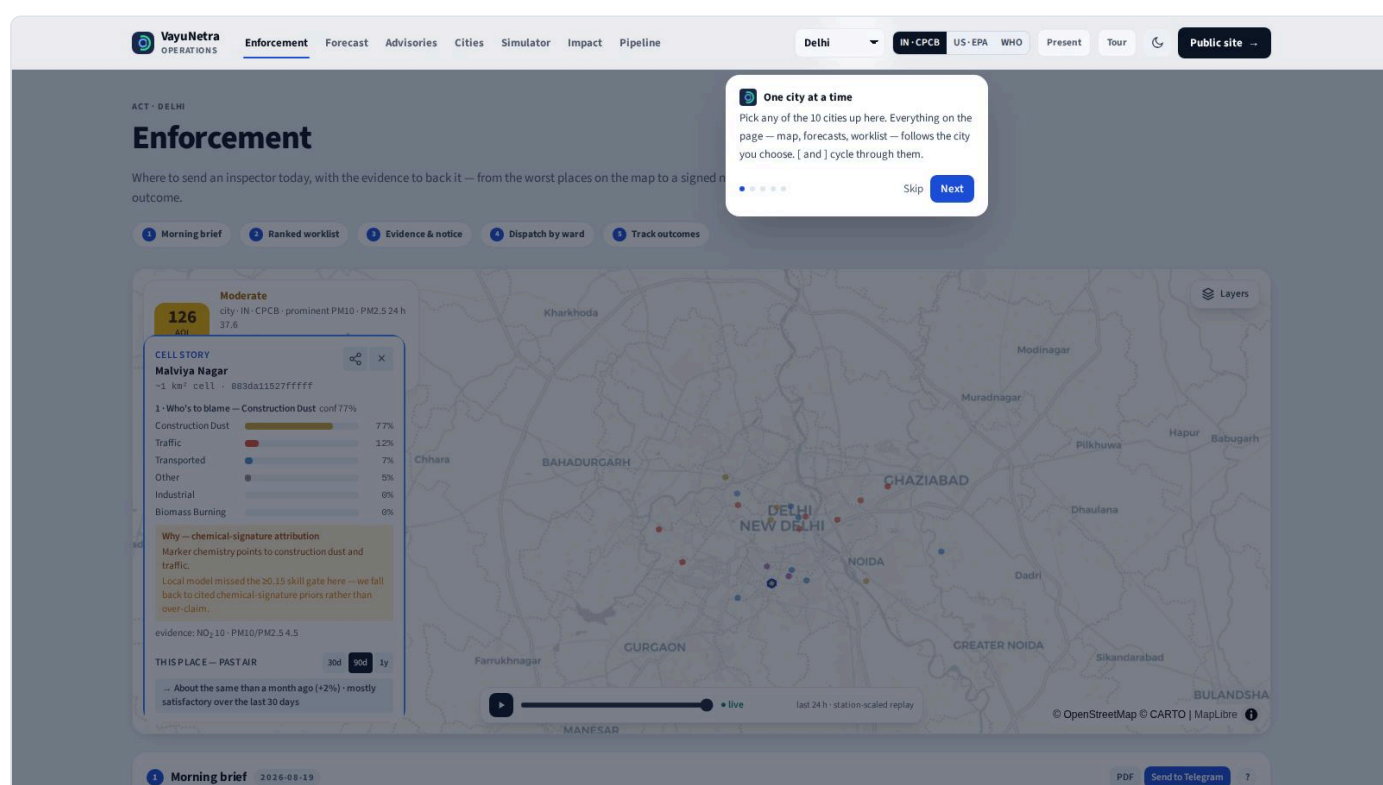
1 Morning brief
2026-08-19
PDF
Send to Telegram
?

Section (key)	One line
Enforcement (1)	Where to send an inspector today, with the evidence — the officer loop. The only section that carries the map.

Section (key)	One line
Forecast (2)	What the air will do in 72 h, how much to trust it, who it affects, what happened before, what is in the air now, and the record behind it
Advisories (3)	What citizens are told, in their language, over which channel — and what today's air does to people
Cities (4)	Ten cities on one scoreboard
Simulator (5)	What-if interventions and the inspector-hour optimiser
Impact (6)	The funding case: ₹, lives, where NCAP money should go, fairness
Pipeline (7)	The six agents, visibly running

Cards are sized to their content: a dense one (the worklist, the outlook) takes the full width; two light ones sit side by side rather than stretching a slider across the page.

4.3 Guided tour (first run)

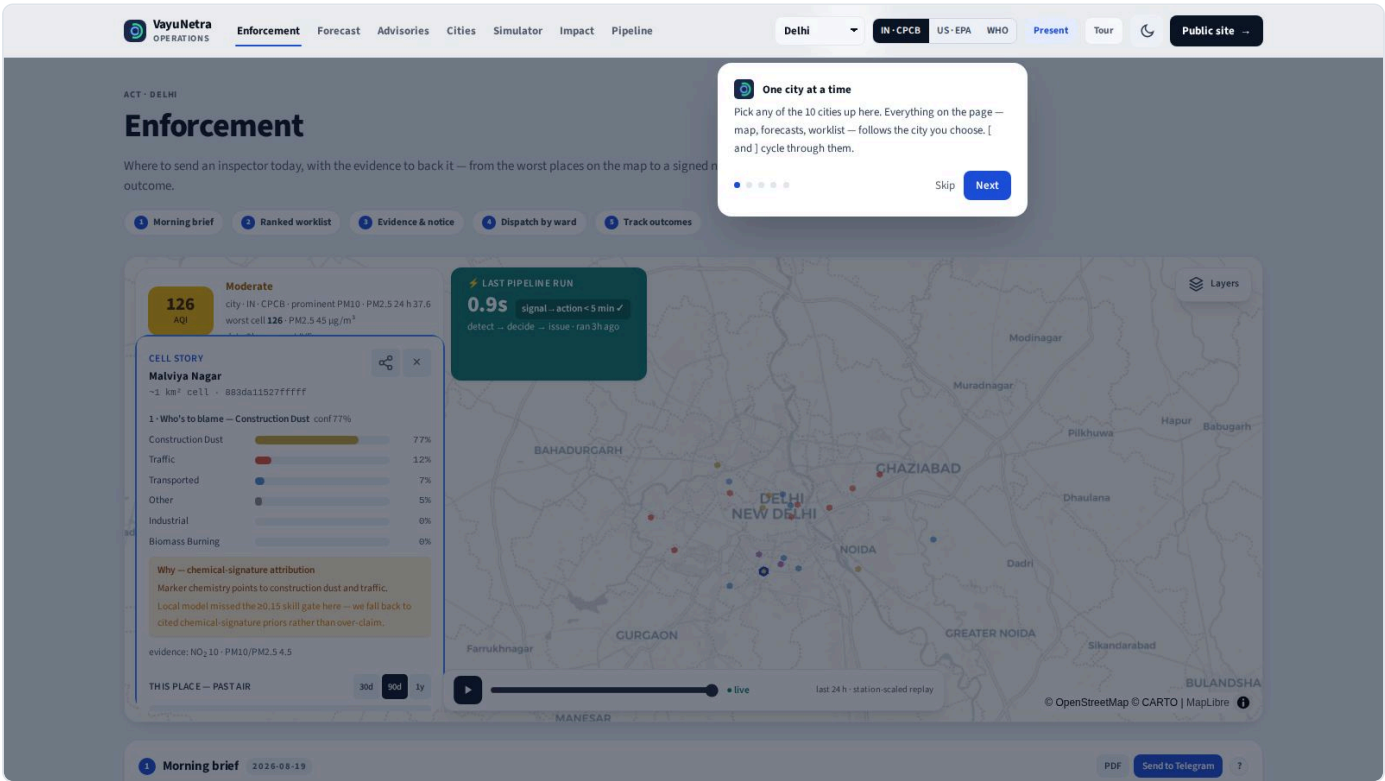


On first visit a five-step tour points at the city selector, the section nav, the map, the spine and the worklist. It shows once (stored in the browser) and replays from **Tour** in the top bar. The tour measures each target and places its card against it, so it stays correct if the layout changes.

4.4 The status strip (on the map)

- **Index tile** — the **city's** index on the chosen scale, with the pollutant setting it, the age of the newest reading and a **LIVE** dot. The worst *cell* is shown beneath it, labelled as such.
- **GRAP Stage · from forecast** — the stage the forecast would trigger (Delhi-NCR only; other cities show their own instrument).
- **Last pipeline run** — the signal → recommendation compute time of the latest agent run.

4.5 Presentation mode

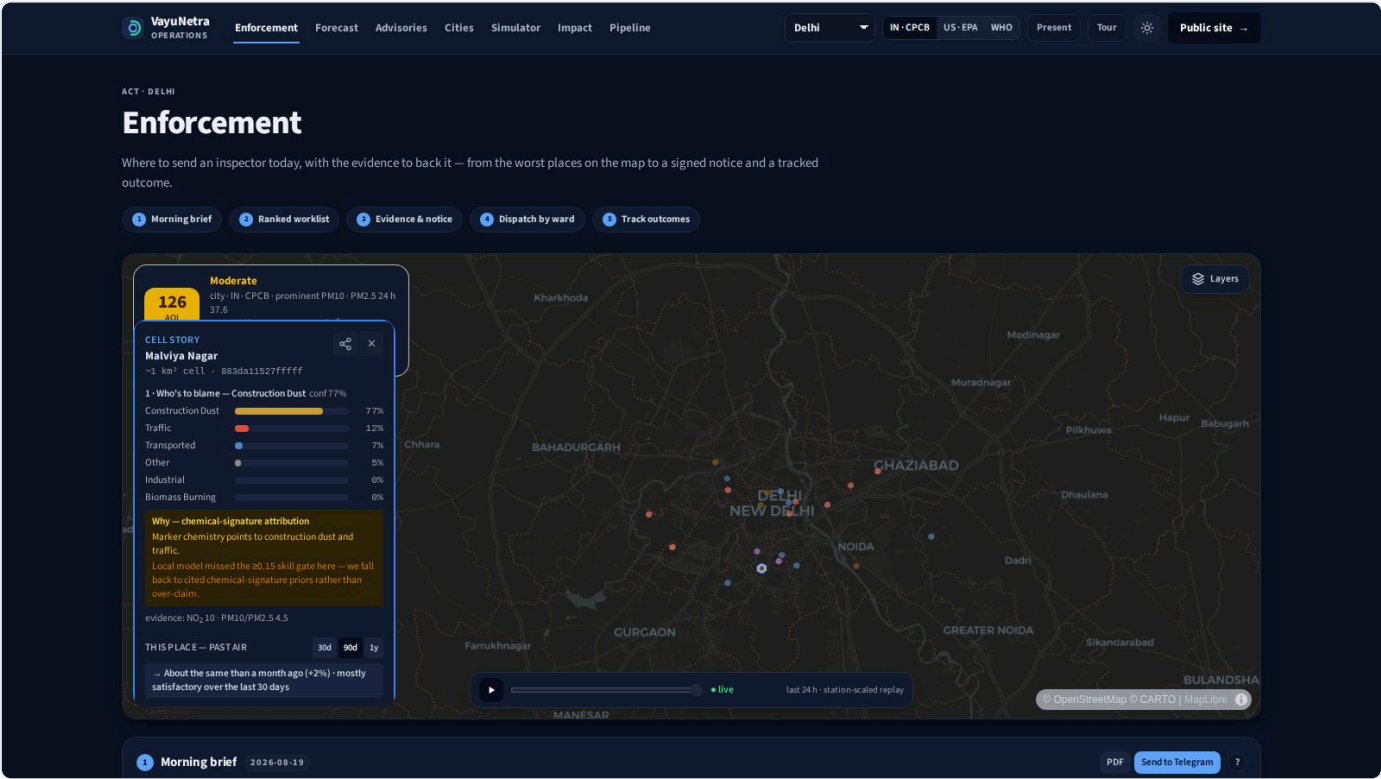
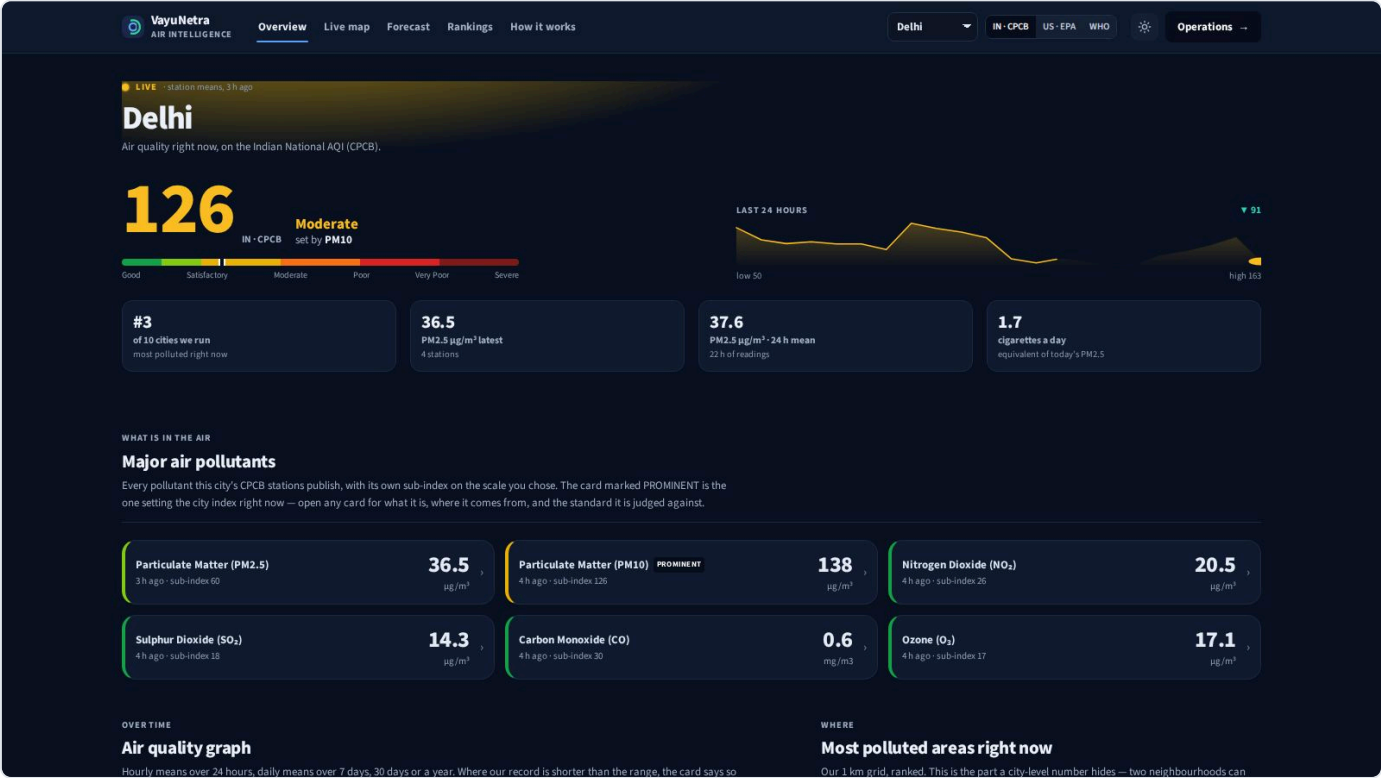


Present (or **P**) enlarges type and spacing for a hall projector without changing content.

4.6 Every screen size, both themes



The whole app is checked from a 320 px phone to a 2560 px monitor, in both themes, by `web/scripts/qa/responsive-audit.mjs` : it computes real WCAG contrast ratios for every text node, flags horizontal overflow, clipped text and touch targets under 44 px, and writes `.qa-out/responsive/report.json` . On a phone the section links move into a menu sheet; nothing scrolls sideways at any width.

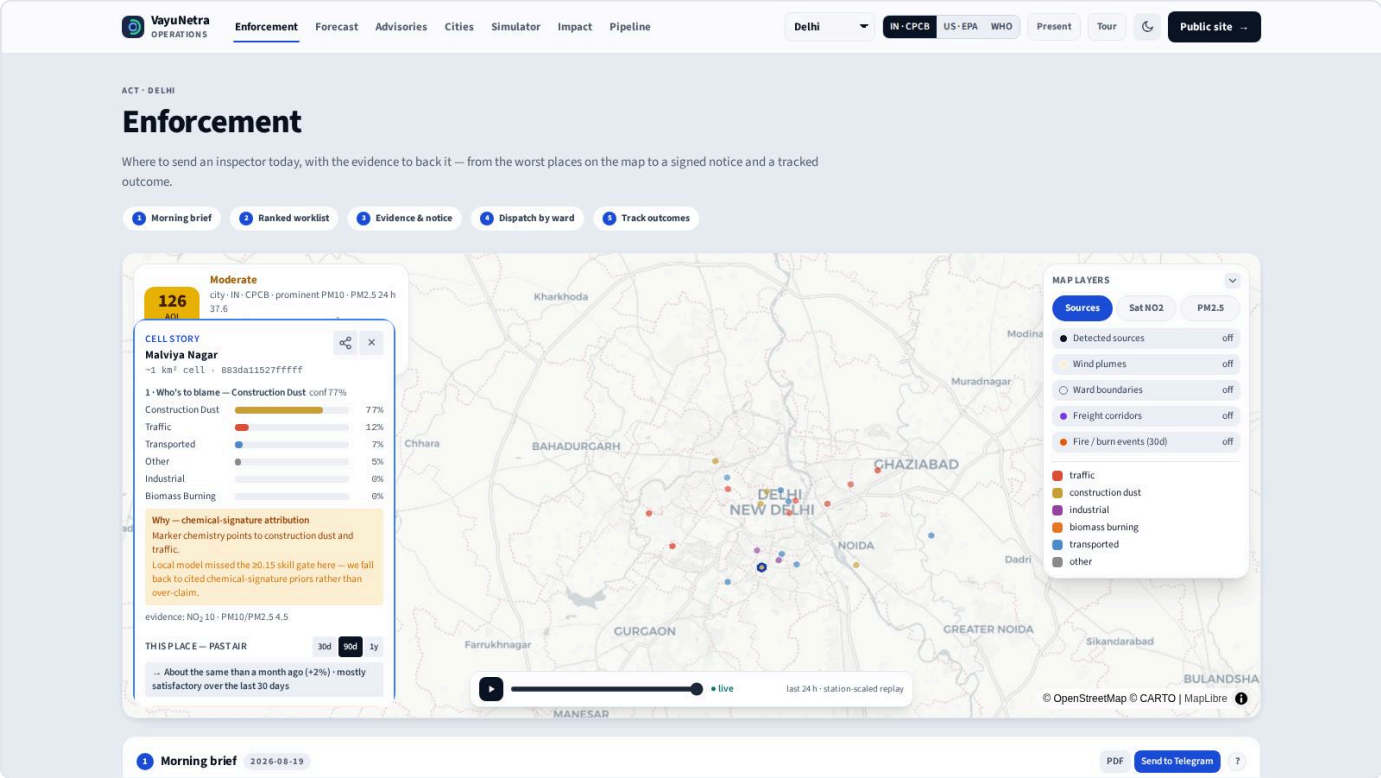


The light/dark switch is in the top bar and is remembered per browser (`?theme=dark` deep-links it). Both themes define the same design tokens, so every surface is correct in either.

5. The map

The map is a MapLibre + deck.gl view of the chosen city; every hexagon is one H3 resolution-8 cell (~1 km²).

5.1 The Layers chip — three view modes



Click **Layers** (top-right of the map) to expand the panel; ▲ collapses it back to a chip.

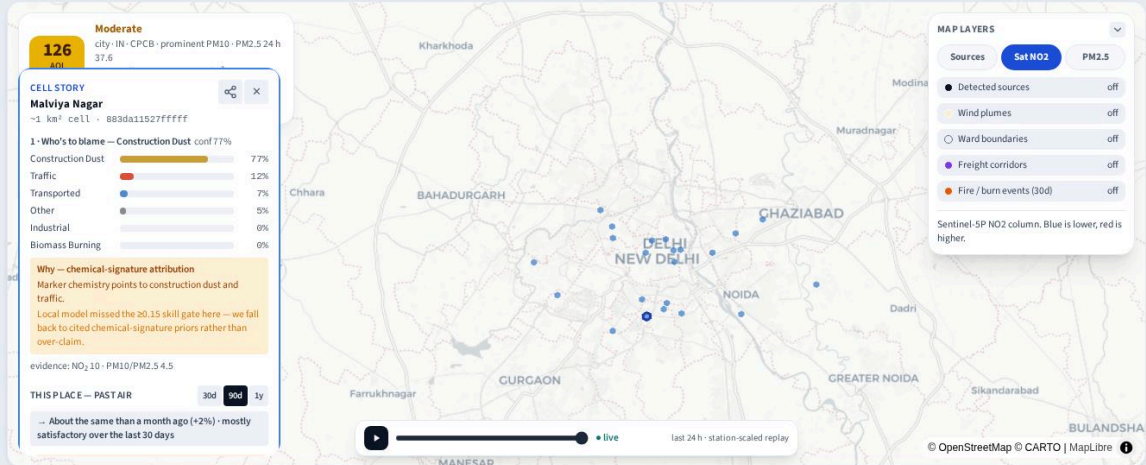
Mode	What the hexagons show
Sources (default)	Who is to blame — each cell filled by its dominant source colour: traffic red · construction dust yellow · industrial purple · biomass burning orange · transported blue · other grey; opacity = dominance. Hover shows the top-3 shares and SHAP drivers.
Sat NO2	Sentinel-5P tropospheric NO ₂ column (blue low → red high) — independent satellite evidence of combustion.
PM2.5	The dense ~1 km model field (E2 downscaler anchored on the live stations). Sub-toggle Stations only ↔ Dense 1 km shows what the model adds beyond the sparse station grid. Legend = CPCB bands.

ACT · DELHI

Enforcement

Where to send an inspector today, with the evidence to back it — from the worst places on the map to a signed notice and a tracked outcome.

- Morning brief
- Ranked workload
- Evidence & notice
- Dispatch by ward
- Track outcomes



1 Morning brief 2026-08-19

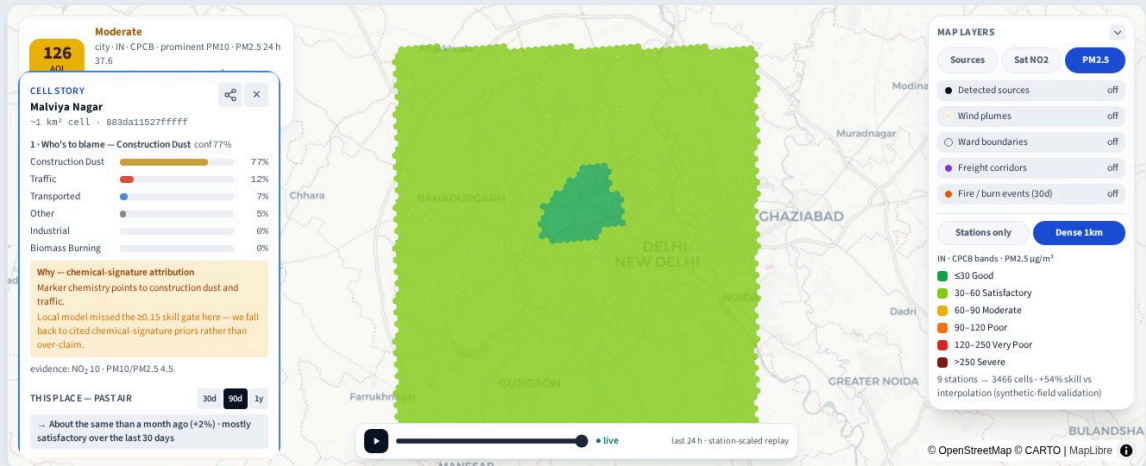
PDF Send to Telegram ?

ACT · DELHI

Enforcement

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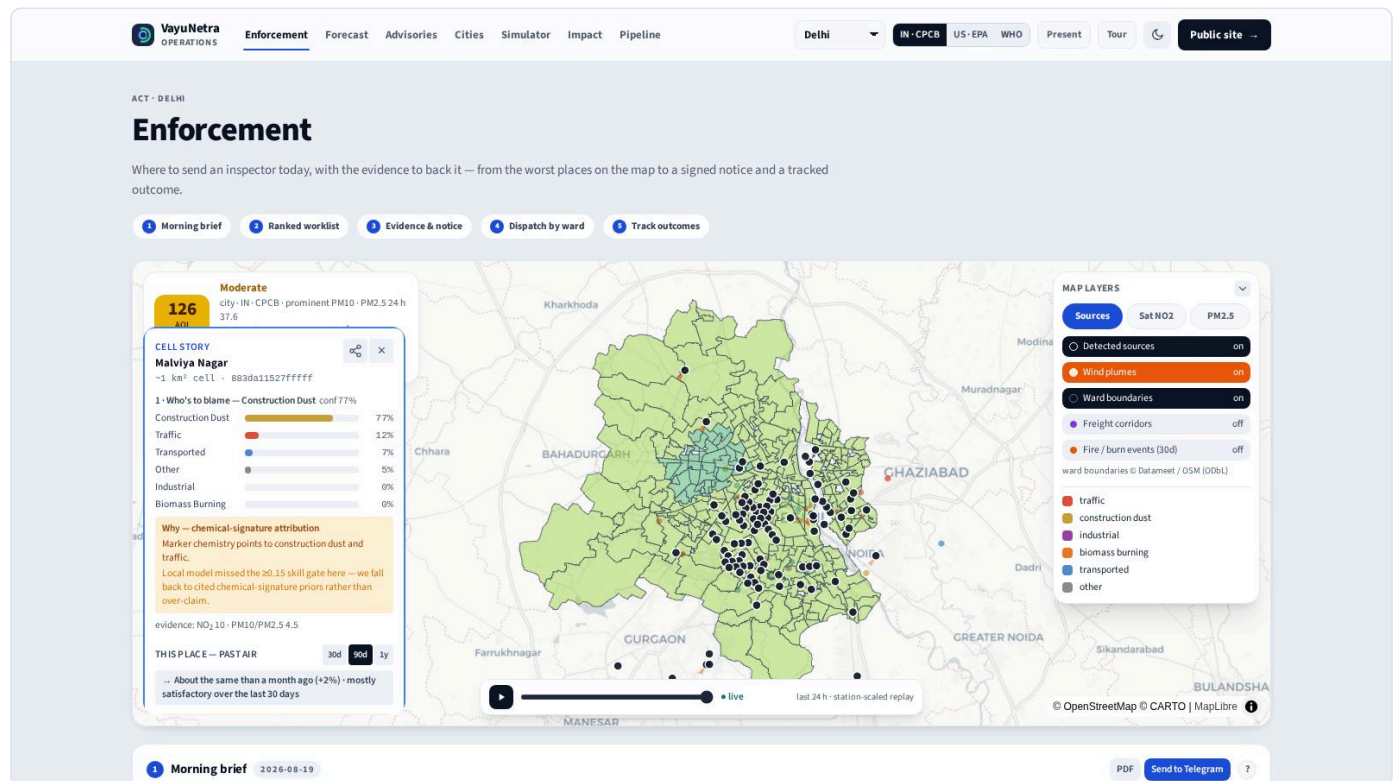
- Morning brief
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1 Morning brief 2026-08-19

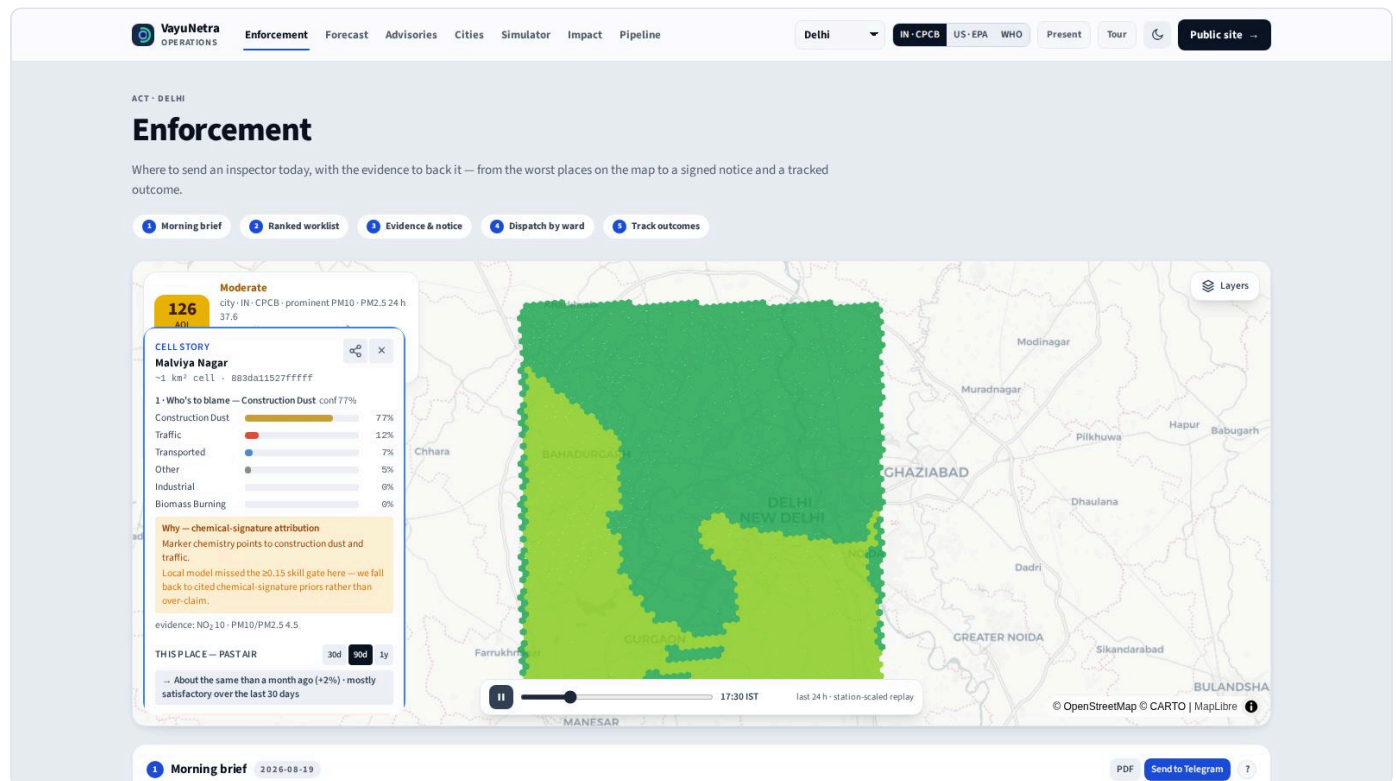
PDF Send to Telegram ?

5.2 Five overlays (independent, all off by default)



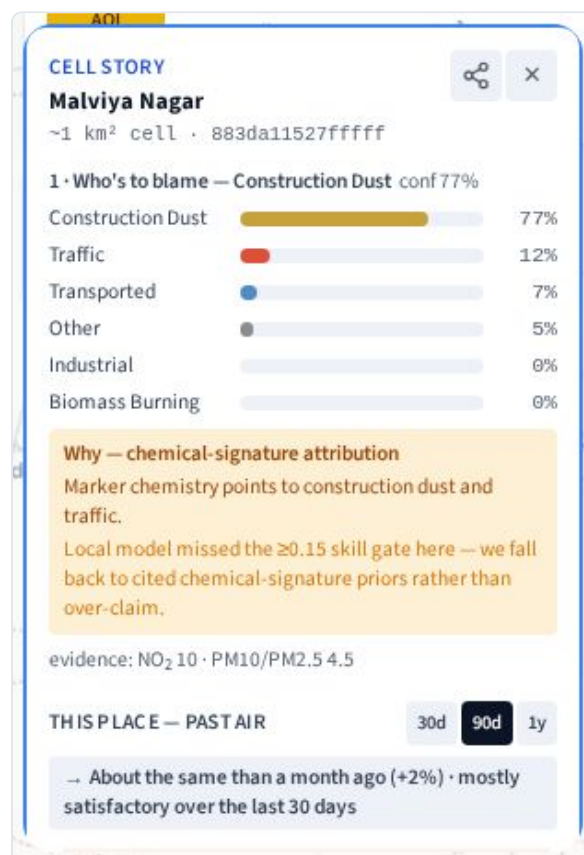
- **Detected sources** — dots for the registered (OSM) and satellite-detected emission sources; **hover a dot to see its real Sentinel-2 image** (sources without one show none — never a stand-in).
- **Wind plumes** — Gaussian-plume footprints from the top sources under the live wind, with a wind-arrow grid (direction; length = speed).
- **Ward boundaries** — administrative wards, each filled by its mean PM2.5 from the dense field (value on hover). The same polygons give every place its name.
- **Freight corridors** — violet truck-route lines with entry-window notes (e.g. Delhi 23:00–07:00).
- **Fire / burn events (30 d)** — NASA FIRMS thermal detections — the stubble / waste-burning layer.

5.3 Play the last 24 hours



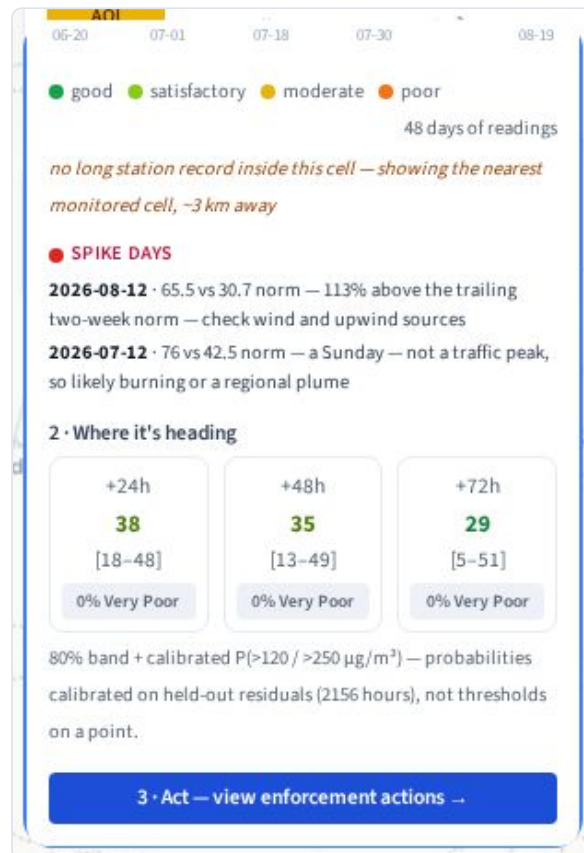
The ► control at the bottom of the map replays the trailing day hour by hour so the city visibly breathes through the evening peak and the morning lull; drag the slider to scrub; it returns to **live** at the end. Labelled honestly: *station-scaled replay* (each cell scaled by its nearest monitor's hourly ratio).

5.4 The Cell Story (click any hexagon)



The heart of the console. It opens on load for the worst cell and for any hexagon you click.

1. **Header** — the place name (ward), the raw H3 id, a **share** button (↗) and close (×).
2. **1 · Who's to blame** — the source-share bars with confidence, then one of two *why* boxes: **green — model attribution**: SHAP drivers in $\mu\text{g}/\text{m}^3$ and the model's own out-of-sample R^2 ("passed the ≥ 0.15 skill gate"); **amber — chemical signature**: where the local model lacks skill the system says so and falls back to cited marker-chemistry priors. It never over-claims.
3. **This place — past air** — daily PM2.5 over 30 d / 90 d / 1 y drawn on the CPCB bands, a one-line verdict, and red **spike-day** markers with a data-backed reason. Cells without a long station record say which monitored cell they are borrowing and how far away it is.
4. **2 · Where it's heading** — the cell's +24/48/72 h PM2.5 with 80 % bands, coloured by AQI band, and the **calibrated probability chips** ("63 % Very Poor+", "91 % Severe").
5. **3 · Act — view enforcement actions** → — jumps to the worklist sorted nearest-to-this-cell first.
6. **Share** — renders a WhatsApp-ready PNG card of this place (blame, verdict, 90-day trend, forecast); uses the phone share sheet, else downloads.



6. Enforcement — the officer loop (section 1)

Where to send an inspector today, with the evidence to back it — from the worst places on the map to a signed notice and a tracked outcome. Steps: **1 Morning brief** · **2 Ranked worklist** · **3 Evidence & notice** · **4 Dispatch by ward** · **5 Track outcomes**.

6.1 Morning brief (step 1)

1

Morning brief

2026-08-19

PDF

Send to Telegram

?

City mean 40 $\mu\text{g}/\text{m}^3$ (Satisfactory) ▲ 10 vs yesterday · worst near Gharoli 45 · 24 h outlook 32.2 (Satisfactory)

WHERE THE AIR IS ABOUT TO TURN
No cell crosses P(Very Poor) $\geq 30\%$ in the 72 h outlook.

TOP ACTIONS TODAY
1. **Construction dust** · Malviya Nagar — 77% of local PM2.5, ~15,000 exposed
2. **Construction dust** · Punjabi Bagh — 60% of local PM2.5, ~15,000 exposed
3. **Construction dust** · Ndmc Charge 3 — 50% of local PM2.5, ~15,000 exposed
+31 more in the worklist below

YESTERDAY'S DISPATCHES
Malviya Nagar: collecting post-dispatch measurements (0.4 d)
Malviya Nagar: collecting post-dispatch measurements (0.4 d)
Punjabi Bagh: collecting post-dispatch measurements (0.6 d)

Advisories: worst tier moderate in 4 zone(s), 2 language(s) · templated from stored rows, no language model

The one page a commissioner reads. Every line is a template over stored rows — no language model: - **Air right now** — city mean PM2.5 (band) vs the same hours yesterday (▲/▼), the worst place, the 24 h outlook; if the last reading is old, its age is printed. - **Where the air is about to turn** — cells whose calibrated P(Very Poor) $\geq 30\%$ at any horizon and that are not already Very Poor, with the horizon and forecast value. - **Top actions today** — the three highest-priority open recommendations (source · place — contribution, residents exposed) with a "+N more" line. - **Yesterday's dispatches** — each dispatched action's measured effect vs the city's drift, or the honest empty state. - **Advisories** — worst tier issued, in how many zones and languages. - **PDF** downloads the brief; **Send to Telegram** pushes it to the city's subscribers (real send; the API reports "skipped" honestly if no bot token is configured). The nightly pipeline sends the brief automatically.

6.2 Enforcement worklist (step 2)

VayuNetra OPERATIONS

Enforcement

Forecast

Advisories

Cities

Simulator

Impact

Pipeline

Delhi

IN · CPCB

US · EPA

WHO

Present

Tour

Public site

(anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).
60% contribution · 15,000 exposed +1 similar site here
Evidence dossier Notice PDF Closed · not applicable · 2026-08-18 History

Construction dust · R. K. Puram

priority 9 · rubric 8/10

Construction dust contributes approximately 51.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #28 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).
51% contribution · 15,000 exposed
Evidence dossier Notice PDF Closed · not applicable · 2026-08-18 History

Construction dust · Ndmc Charge 3

priority 7 · rubric 8/10

Construction dust contributes approximately 49.5% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #17 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).
50% contribution · 15,000 exposed +12 similar sites here
Evidence dossier Notice PDF Approve Dismiss

Construction dust · near Madanpur Khadar

priority 5 · rubric 8/10

Construction dust contributes approximately 45.3% of PM2.5 in this cell (~12,338 residents exposed); Tajpur Ecopark is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).
45% contribution · 12,338 exposed
Evidence dossier Notice PDF Approve Dismiss

Construction dust · Ndmc Charge 3

priority 3 · rubric 8/10

Construction dust contributes approximately 32.9% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #12 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).
33% contribution · 15,000 exposed +3 similar sites here
Evidence dossier Notice PDF Closed · not applicable · 2026-08-18 History

Industrial emissions · Punjabi Bagh

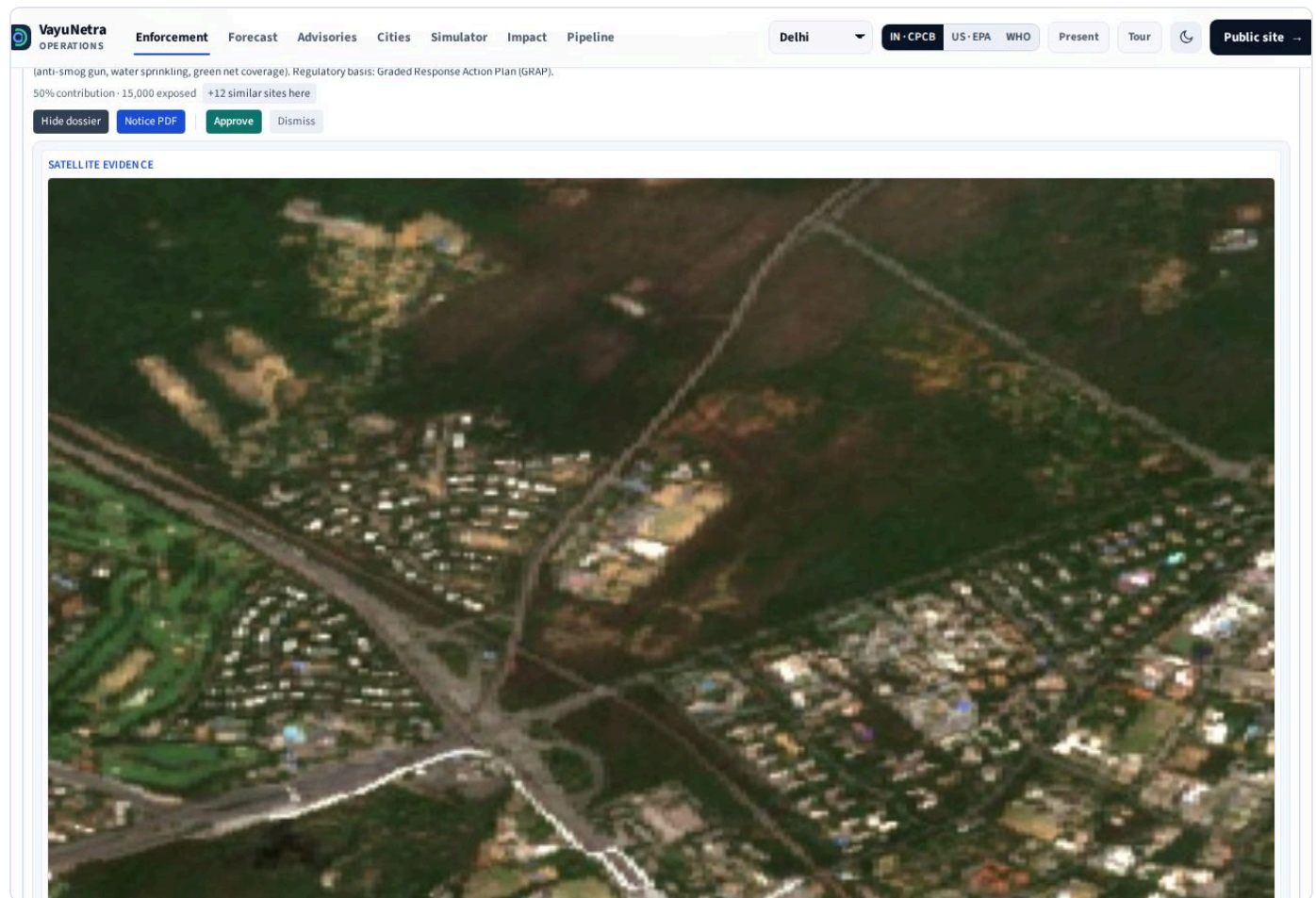
priority 3 · rubric 7/10

Industrial emissions contributes approximately 22.5% of PM2.5 in this cell (~11,854 residents exposed); Rithala Power Plant is the registered industry source at this location. Industrial inspection required: verify stack emission norms, Consent-to-Operate (CTO) compliance, and OCEMS data. Regulatory basis: Graded Response Action Plan (GRAP).
23% contribution · 11,854 exposed
Evidence dossier Notice PDF Approve Dismiss

Ranked recommendations, each titled **source type · place** with its priority and rubric score. Filters **All / Construction / Industrial / Waste**, free-text **search**, and the **acting as** field (the officer name stamped on every action; the demo has no sign-in — a deployment binds this to the authenticated user). If a cell is focused, sorting switches to *nearest first* (badge + "~N km" / "📍 this cell" chips). Identical detections in one cell collapse into one card ("~N similar sites here").

Each card: the rationale (what the source is, its % contribution, residents exposed, what to inspect, the regulatory basis), then the buttons **Evidence dossier** · **Notice PDF** · **Approve** · **Dismiss**.

6.3 Evidence dossier and notice (step 3)



- **Evidence dossier** expands in place: the **Sentinel-2 satellite patch** of the site (with detection confidence) and up to three **regulatory citations** retrieved by RAG from the NCAP / GRAP / CPCB / Air-Act corpus (GRAP/CAQM only where they legally bind — Delhi-NCR; CPCB dust norms and NCAP elsewhere), each with a match score.
- **Notice PDF** downloads a draft enforcement notice: addressee, cited provisions, 24-hour IST deadline, the satellite image, a **projected impact of compliance** chart (forecast with vs without this source's share at +24/48/72 h — a screening estimate), the issuing authority, and a **PROVENANCE** block. Stamped *DRAFT — pending officer authorisation*; the system never auto-sends. Live assembly takes 5–12 s.

6.4 Approve → Dispatch → Close → History



- **Approve** moves the item to its ward's field queue (step 4). **Dismiss** removes it.
- **Dispatch team** freezes the cell's 7-day PM2.5 baseline and **arms before/after tracking** (step 5); the card shows *Dispatched · tracking armed*.

Construction dust · Ndmic Charge 3

priority 7 · rubric 8/10

Construction dust contributes approximately 49.5% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #17 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

50% contribution · 15,000 exposed +12 similar sites here

Evidence dossier

Notice PDF

✓ Dispatched · tracking armed

Close case

History

Field finding

violation found

note (optional): what was seen, notice served, follow-up...

Record & close

- Close case** records the field finding — *violation found · compliant · site inaccessible · not applicable* — and an optional note; **Record & close** stores it and the card shows *Closed · finding · date*.

Construction dust · Ndmic Charge 3

priority 7 · rubric 8/10

Construction dust contributes approximately 49.5% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #17 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

50% contribution · 15,000 exposed +12 similar sites here

Evidence dossier

Notice PDF

✓ Closed · not applicable · 2026-08-19

Hide history

19 Aug, 12:08 pm proposed → **approved** by Guide walkthrough

19 Aug, 12:08 pm approved → **dispatched** by Guide walkthrough

19 Aug, 12:08 pm dispatched → **closed** by Guide walkthrough not applicable "user-guide walkthrough — no field visit"

- History** shows the audit trail: from → to, who (acting as), when, finding, note. It is append-only (`enforcement_status_log`), survives the nightly run — which refreshes evidence and priority *in place* for acted-upon items and replaces only still-proposed ones — and survives even deletion of the record.

6.5 Ward dispatch queues (step 4)

4

Ward dispatch queues

EMPTY

?

Nothing dispatched yet. Approve or dispatch a worklist item above and it lands in its ward's field queue here.

Approved and dispatched recommendations group into per-ward field queues (the grid-supervision pattern); wards are resolved offline from the shipped boundary polygons. Before anything is dispatched the card says so plainly.

6.6 Intervention tracking and PRANA export (step 5)

Each dispatched action's measured effect: the cell's PM2.5 after dispatch minus its frozen baseline, corrected for the city-wide drift over the same window (marked *provisional* for the first 7 days). **Export as NCAP action-plan evidence** downloads the PRANA-ready CSV — every tracked action with baseline / after / effect / status / closure finding, mapped to the NCAP spending head the city reports against.

6.7 City Intel

City Intel

registry sources

103

traffic index

60

sensitive zones

1000

1,542 hospitals/clinics · 1,485 schools mapped — advisories escalate where forecast air is bad and sensitive people are.

TOP REGISTERED SOURCES

Satellite-detected construction site #12

74%

Satellite-detected construction site #13

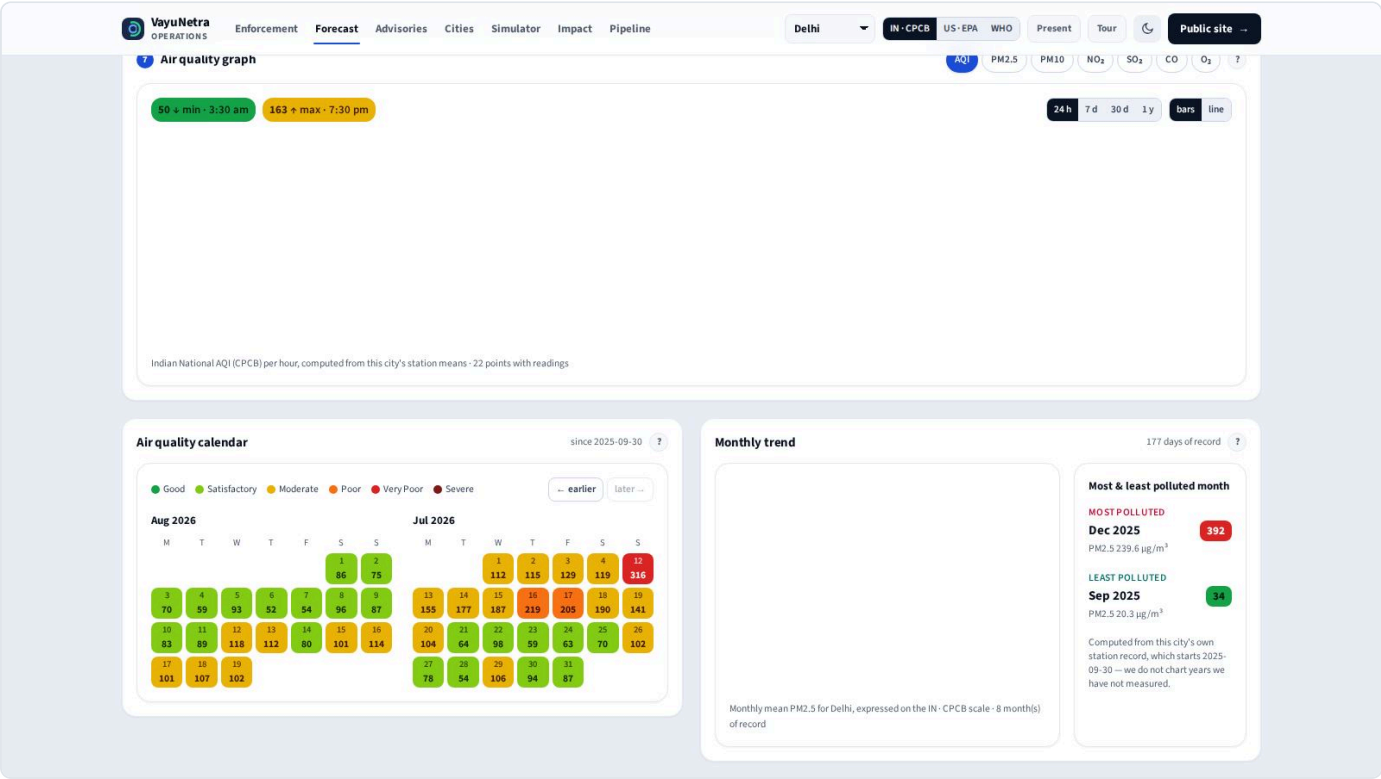
87%

Registry-source count, traffic index, sensitive-zone count, hospitals / schools, top registered sources — the static picture behind the live one.

7. Forecast (section 2)

What the air will do in the next 72 hours, how much to trust that, who it will affect — and what happened before. Steps: 1 72-hour outlook · 2 How good is it, really · 3 Real orders, in hindsight · 4 Who is in the forecast · 5 The past.

7.1 72-hour outlook



Horizon tabs +24h / +48h / +72h; the city-average forecast line with the shaded 80 % band and the grey dashed persistence baseline ("tomorrow = today" — the bar to beat). The **measured skill note** under the chart is read from the benchmark artifact, never typed in. **Spike alerts** list cells forecast $\geq 90 \mu\text{g}/\text{m}^3$ (worst first). Model: LightGBM quantile regression, retrained daily per city on the trailing 90 days, blended with persistence, conformally calibrated; every cell carries $P(>120)$ and $P(>250)$.

7.2 Forecast validation (step 2)

2

Forecast validation

MEASURED

multi-season

live 90d

?

Strict temporal split: trained on 2025-02-17 → 2025-11-01, tested on 2025-11-01 → 2026-08-15 (39 station cells, 234,491 test hours @24h). The served forecast (LightGBM median blended with persistence, weight from the calibration tail) vs persistence, weekly seasonal-naive and hour-of-day climatology on one shared support mask.

horizon	vs persistence	vs seasonal	winter	>120 hours	80% PI
+24h	+9%	+26%	+7%	+7%	+78%
+48h	+13%	+20%	+11%	+11%	+78%
+72h	+12%	+16%	+11%	+10%	+77%

Early warning on Very Poor onsets (clean at issue time, >120 µg/m³ at t+h): alarming on the calibrated probability ($P \geq 30\%$) flags 54% @24h · 54% @48h · 51% @72h of onsets (precision 68% · 66% · 64%); the plain median alarm 24% · 31% · 27%. Persistence catches 0% by construction — a "tomorrow = today" system can never warn of a spike before it starts.

Recomputed 2026-08-17 · negative numbers are kept, not hidden · full tables in docs/benchmarks/delhi.md

How today's attribution was made

22 cells

signature priors 22

no cell passed the $R^2 \geq 0.15$ gate · mean confidence 0.54 · cells reporting NO₂/CO/SO₂ in the last 24 h 1

recomputed daily from the latest hour; the split moves with station marker coverage — stated, not smoothed

The strict temporal-split benchmark: skill vs persistence and weekly seasonal-naive per horizon, winter-only and Very-Poor-hours slices, 80 % interval coverage, and the **early-warning line** — how many clean→Very-Poor onsets the alarm on $P \geq 0.3$ flags 1–3 days ahead (persistence = 0 by construction). Toggle **multi-season / live 90d**. Negative numbers stay visible. Below it, **How today's attribution was made** — cells per method (per-cell model · shrunk to the city model mean · signature priors), median R^2 , mean confidence, cells with a gas marker in the last 24 h. Sources: docs/BENCHMARKS.md , docs/EARLY_WARNING.md , GET /metrics/benchmark , GET /metrics/attribution .

7.3 Real interventions, in hindsight (step 3)

3

Real interventions, in hinds...

WINTER 2025-26

would we have warned?

did the air change?

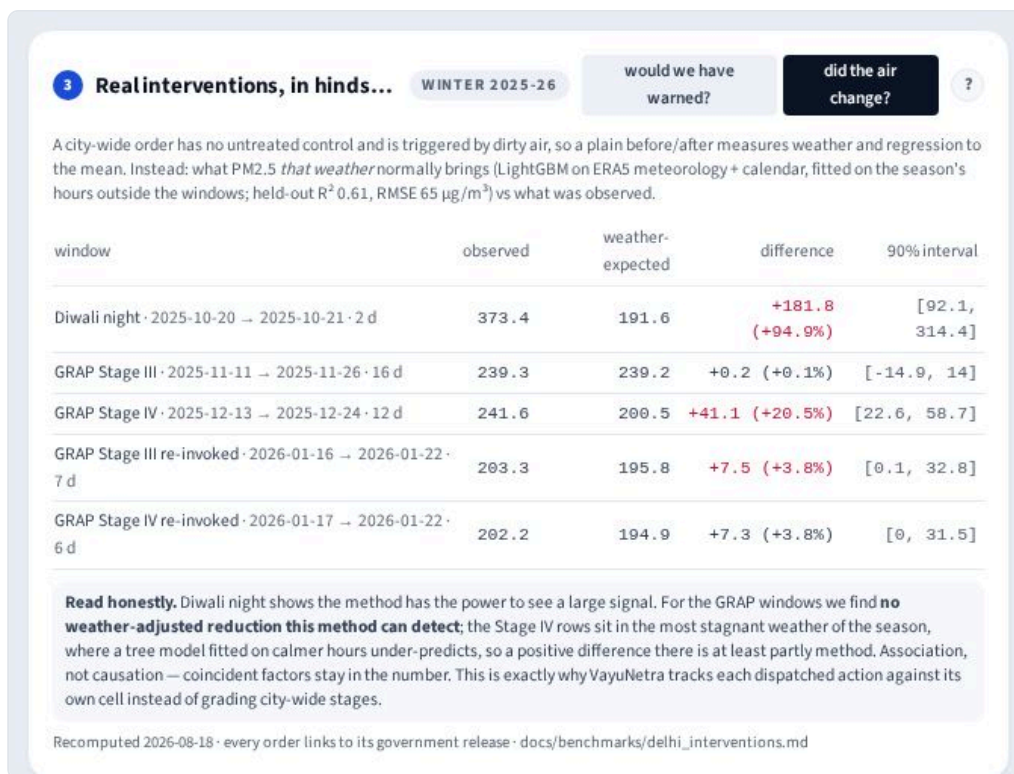
?

CAQM's GRAP orders for Delhi-NCR, winter 2025-26 (dates from the government releases), against the served forecast replayed on 39 station cells: the calibrated $P(>120)$ the system carried **24 / 48 / 72 h before each order**, and the share of cells past the $P \geq 30\%$ alarm.

order	at order	$P(>120)$ 24 h before	48 h	72 h	cells alarming
GRAP Stage I · 2025-10-14	83 . 8	1%	2%	3%	0% of 38
GRAP Stage II · 2025-10-19	164 . 5	9%	6%	7%	1% of 38
GRAP Stage III · 2025-11-11 low coverage	325 . 6	94%	76%	91%	100% of 2
GRAP Stage IV · 2025-12-13	381 . 8	83%	42%	33%	99% of 38
GRAP Stage III re-invoked · 2026-01-16 low coverage	213 . 8	85%	72%	57%	100% of 1
GRAP Stage IV re-invoked · 2026-01-17 low coverage	332 . 7	72%	83%	72%	100% of 1

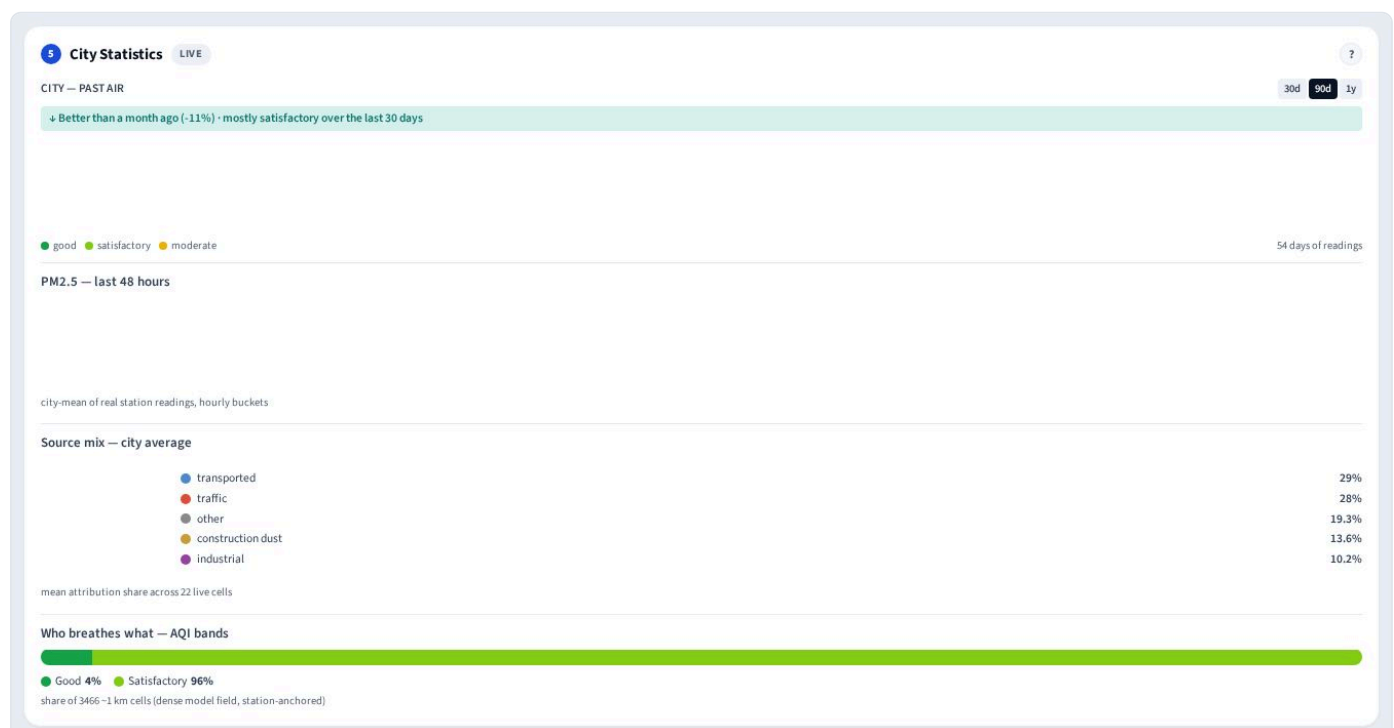
Read honestly. The Stage III (11 Nov) and Stage IV (13 Dec) escalations were flagged a day ahead across the whole station network; the two October orders were not — the mid-October rise came faster than the model saw. Persistence, by construction, only ever repeats today's level. Status quo, for contrast: 13 of 17 GRAP orders in winter 2025-26 were passed after the AQI had already crossed that stage's threshold (press analysis, not an audit figure).

Recomputed 2026-08-18 · every order links to its government release · docs/benchmarks/delhi_interventions.md



The CAQM GRAP escalations of winter 2025-26 (and Diwali night), dated from government releases (each order links to its source), replayed against the served forecast: **would we have warned?** shows P(>120) 24/48/72 h before each order and the share of station cells past the alarm (low-coverage weeks flagged); **did the air change?** shows observed vs weather-expected PM_{2.5} inside each window with a bootstrap interval — where the honest answer is "no detectable change", it says no. docs/OUTCOMES.md , GET /metrics/interventions .

7.4 City Statistics (steps 4–5)



- **Who is in the forecast?** — expected people in Very Poor / Severe air at +24/48/72 h = Σ cell population $\times$ calibrated P(> band); GPW v4.11 population where sampled, cited city population otherwise; exposure, not mortality (docs/HEALTH_IMPACT.md).
- **City — past air** — daily PM_{2.5} for the whole city over 30 d / 90 d / 1 y with verdict and spike days (raw readings u the archived daily rollout — Delhi's 1-year view shows the real winter).

- **Last 48 hours** — hourly station means; the live **source-mix donut**; **Who breathes what** — share of cells in each CPCB band.

8. Advisories — the citizen loop (section 3)

Inform: what citizens are told, in their language, over which channel. Steps: **1 Advisories by ward** • **2 Send it** • **3 Clean-air routes** • **4 Citizen reports.**

8.1 Citizen Advisory

VayuNetra OPERATIONS Enforcement Forecast **Advisories** Cities Simulator Impact Pipeline

Delhi IN - CPCB US - EPA WHO Present Tour Public site

What today's air does to people NOT MEDICAL ADVICE

1.7 cigarettes / day

Breathing Delhi's air over the last 24 hours is comparable to smoking about **1.7** cigarettes a day.

weekly **11.9** monthly **51**

Basis: 24-h mean PM_{2.5} 37.6 µg/m³. Communication device, not a clinical dose: it converts the last 24 h mean PM_{2.5} assuming continuous exposure. Source: Berkeley Earth, Air Pollution and Cigarette Equivalence (2015).

What to do now — air is Moderate

Breathing discomfort for people with asthma, and for those with heart or lung disease; the general public is largely unaffected.

Heavy exertion
Reduce if sensitive

N95 mask
If sensitive

Windows
Open away from traffic

Air purifier
Optional

Understand your risk

Asthma Heart conditions Allergies Sinus Cold / flu Chronic (COPD)

Asthma

Risk today is **mild** — the air is Moderate. Wheezing, shortness of breath, chest tightness and cough become more frequent as PM_{2.5} rises.

DO

- ✓ Keep your reliever inhaler with you and use it as prescribed.
- ✓ Move exercise indoors or to the cleanest hours (usually early afternoon).
- ✓ Run an air purifier in the room you sleep in and keep windows shut at peak hours.

DON'T

- ✗ Don't exercise outdoors without a mask when the air is Poor or worse.
- ✗ Don't stay near idling traffic, burning waste or construction dust.
- ✗ Don't skip preventive medication because you feel fine today.

General public health guidance for the current air quality, not medical advice. If you have symptoms, consult a doctor. Sources: CPCB, National Air Quality Index (2014) — associated health impacts table; WHO Global Air Quality Guidelines (2021); Berkeley Earth, Air Pollution and Cigarette Equivalence (2015).

Citizen Advisory Hindi ?

App Telegram IVR call Big screen

how the same advisory reaches citizens on each channel

zone-0e3d moderate

Delhi zone-0e3d: अगले 24 घंटों में हवा मध्यम रहने का अनुमान है. संवेदनशील लोग (बच्चे, बुजुर्ग, दमा रोगी) लंबी बाहरी मेहनत कम करें.

zone-00b3 moderate

Delhi zone-00b3: अगले 24 घंटों में हवा मध्यम रहने का अनुमान है. संवेदनशील लोग (बच्चे, बुजुर्ग, दमा रोगी) लंबी बाहरी मेहनत कम करें.

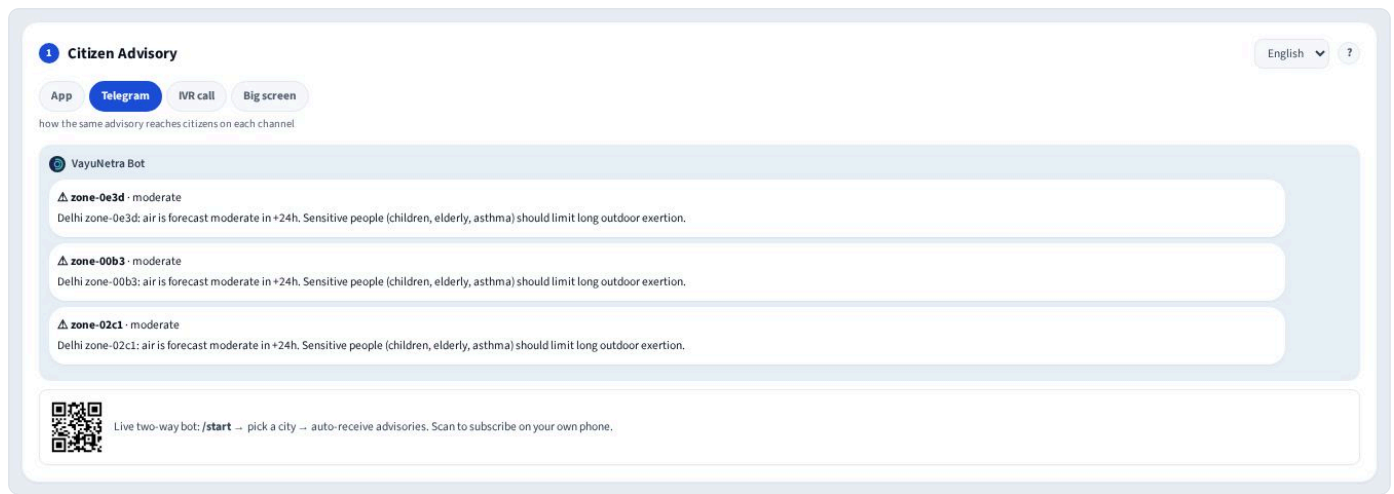
zone-02c1 moderate

Delhi zone-02c1: अगले 24 घंटों में हवा मध्यम रहने का अनुमान है. संवेदनशील लोग (बच्चे, बुजुर्ग, दमा रोगी) लंबी बाहरी मेहनत कम करें.

zone-02c3 moderate

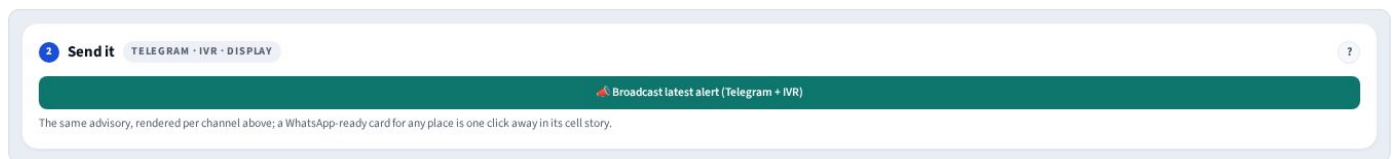
Delhi zone-02c3: अगले 24 घंटों में हवा मध्यम रहने का अनुमान है. संवेदनशील लोग (बच्चे, बुजुर्ग, दमा रोगी) लंबी बाहरी मेहनत कम करें.

Ward-level advisories tiered by forecast risk and vulnerability. **Language dropdown:** the city's showcase language first (Hindi in Delhi, Marathi in Mumbai/Pune, Kannada in Bengaluru...), then the others — full native-script text from deterministic templates (LLM-free by design, so medical advice cannot be hallucinated; script-validated in code; native-speaker review status per language in `docs/ADVISORY_REVIEW.md`). **Channel tabs** show how the same advisory renders on each channel:



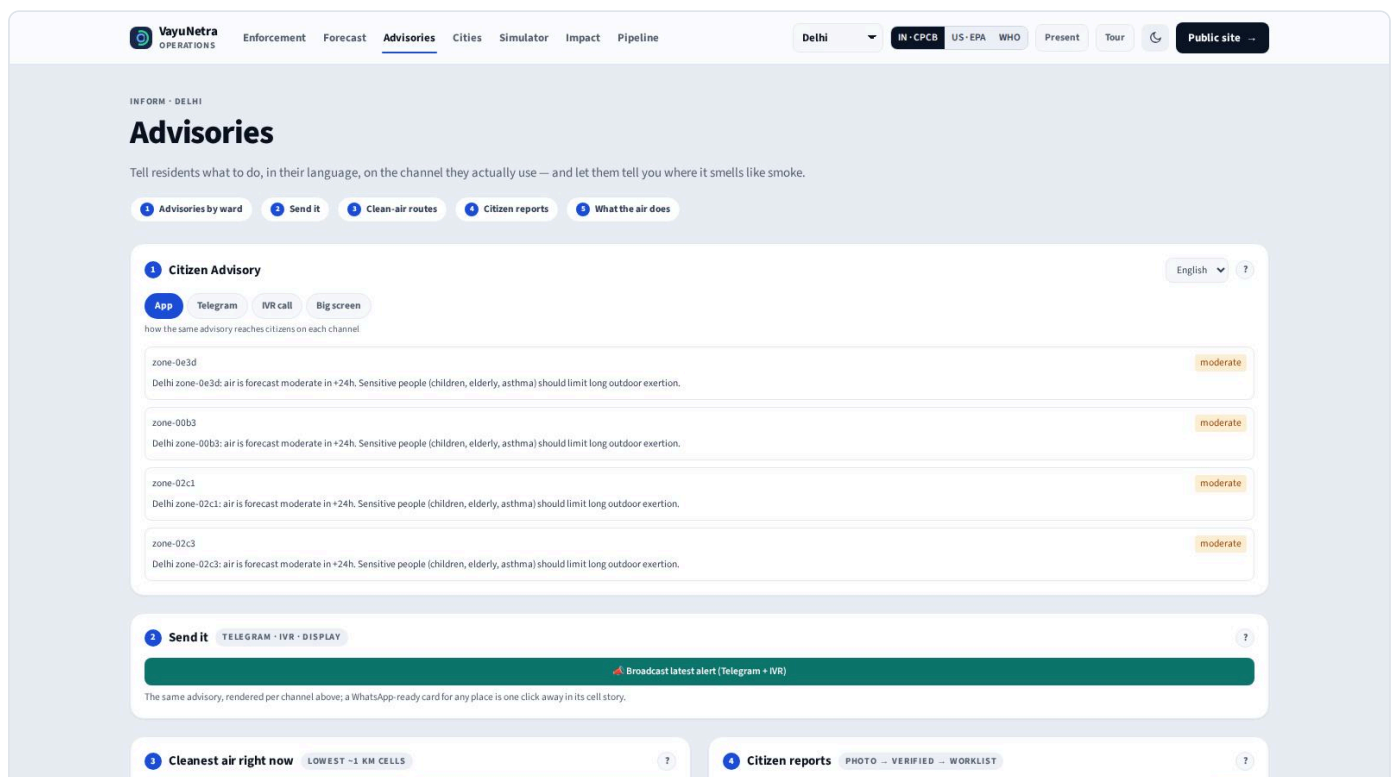
- **App** — advisory cards per ward with risk-tier badges.
- **Telegram** — the chat rendering + QR; the real bot **@aqivayu_bot** is live: `/start` → pick a city → receive advisories.
- **IVR call** — the spoken script, plus the call-in line: a keypad menu of the ten cities reads that city's latest advisory — Hindi voice for Hindi-first cities, English elsewhere (no Polly voice exists yet for the other six scripts; those calls read English — stated on screen).
- **Big screen** — high-contrast public-display rendering.

8.2 Send it



Broadcast latest alert (Telegram + IVR) sends a real Telegram message and places a real call — behind an are-you-sure confirmation and a server-side rate limit. Share cards for WhatsApp come from the Cell Story.

8.3 Cleanest air right now (step 3)



The four lowest-PM2.5 ~1 km zones from the dense field, each with **Directions** ↗ (Google Maps) and a corridor exposure screen from the city centre — labelled a modelled guide, not a measurement.

8.4 Citizen reports (step 4)

The screenshot displays the VayuNetra OPERATIONS interface. The top navigation bar includes links for Enforcement, Forecast, Advisories (active), Cities, Simulator, Impact, and Pipeline. A location dropdown is set to Delhi, and standard agency logos (IN-CPCB, US-EPA, WHO) and utility buttons (Present, Tour, Public site) are visible. The main header reads 'INFORM · DELHI' followed by the 'Advisories' title. A sub-header states: 'Tell residents what to do, in their language, on the channel they actually use — and let them tell you where it smells like smoke.' Below this is a progress bar with five steps: 1. Advisories by ward, 2. Send it, 3. Clean-air routes, 4. Citizen reports (current step), and 5. What the air does. The 'Citizen Advisory' section shows four zones (0e3d, 00b3, 02c1, 02c3) with a 'moderate' air quality forecast and a text advisory for sensitive groups. The 'Send it' section shows a green broadcast button for Telegram and IVR. The bottom navigation bar shows 'Cleanest air right now' and 'Citizen reports' (active).

Report a pollution source: photo + category + location → a public list with a 72-hour SLA clock (received → verified → actioned → resolved / rejected; breaches in red). An officer marking a report *verified* registers the location as a candidate source so the next enforcement run scores it — the loop closes from the citizen side.

9. Cities (section 4)

[Enforcement](#)
[Forecast](#)
[Advisories](#)
[Cities](#)
[Simulator](#)
[Impact](#)
[Pipeline](#)

[Delhi](#)
[IN - CPCB](#)
[US - EPA](#)
[WHO](#)
[Present](#)
[Tour](#)
[Public site](#)

COMPARE · DELHI

Cities

Ten cities on one scoreboard: who is worse, what is driving it, and which playbook worked elsewhere.

1 Scoreboard

2 What worked elsewhere

1 Multi-City Compare

7

Kolkata: construction dust · Bengaluru: industrial · Mumbai: construction dust · Delhi: transported · Pune: construction dust · Lucknow: industrial · Chennai: construction dust · Ahmedabad: transported · Hyderabad: industrial · Jaipur: transported · city-average PM2.5

+24h

avg now

CLEAN·AIR·RAN·KING — Swachh Yayu style, by current PM2.5 (cleanest first)

#1 Mumbai

27

avg 16 $\mu\text{g}/\text{m}^3$

Construction Dust

~40,168 deaths/yr · ₹2.01 lakh cr/yr

-- pre-wet exposed soil

Compliance: 172 recommendations — all awaiting officer review

+24h 12 $\mu\text{g}/\text{m}^3$

construction-winter

stable

#2 Pune

32

avg 19 $\mu\text{g}/\text{m}^3$

Construction Dust

~12,693 deaths/yr · ₹63,466 cr/yr

-- pre-wet exposed soil

Compliance: 14 recommendations — all awaiting officer review

+24h 16 $\mu\text{g}/\text{m}^3$

construction-winter

stable

#3 Bengaluru

32

avg 19 $\mu\text{g}/\text{m}^3$

Industrial

~19,866 deaths/yr · ₹99,332 cr/yr

-- verify consent-to-operate limits

Compliance: 120 recommendations — all awaiting officer review

+24h 11 $\mu\text{g}/\text{m}^3$

industrial-signature

stable

#4 Chennai

33

avg 20 $\mu\text{g}/\text{m}^3$

Construction Dust

~15,838 deaths/yr · ₹79,191 cr/yr

-- pre-wet exposed soil

Compliance: 11 recommendations — all awaiting officer review

+24h 21 $\mu\text{g}/\text{m}^3$

construction-winter

stable

#5 Lucknow

36

avg 22 $\mu\text{g}/\text{m}^3$

Industrial

~10,856 deaths/yr · ₹54,278 cr/yr

-- verify consent-to-operate limits

Compliance: 6 recommendations — all awaiting officer review

+24h 19 $\mu\text{g}/\text{m}^3$

industrial-signature

stable

#6 Hyderabad

36

avg 22 $\mu\text{g}/\text{m}^3$

Industrial

~18,663 deaths/yr · ₹93,316 cr/yr

-- verify consent-to-operate limits

Compliance: 15 recommendations — all awaiting officer review

+24h 19 $\mu\text{g}/\text{m}^3$

industrial-signature

stable

#7 Kolkata

62

avg 25 $\mu\text{g}/\text{m}^3$

Construction Dust

~34,320 deaths/yr · ₹1.72 lakh cr/yr

-- pre-wet exposed soil

Compliance: 14 recommendations — all awaiting officer review

+24h 23 $\mu\text{g}/\text{m}^3$

construction-winter

stable

#8 Delhi

56

avg 34 $\mu\text{g}/\text{m}^3$

Transported

~73,395 deaths/yr · ₹3.67 lakh cr/yr highest burden

-- maintain monitoring

Compliance: 45 recommendations - 1 dismissed

+24h 32 $\mu\text{g}/\text{m}^3$

transported-signature

stable

#9 Ahmedabad

58

avg 35 $\mu\text{g}/\text{m}^3$

Transported

~19,013 deaths/yr · ₹95,067 cr/yr

-- maintain monitoring

Compliance: 11 recommendations — all awaiting officer review

+24h 26 $\mu\text{g}/\text{m}^3$

transported-signature

stable

#10 Jaipur

65

avg 39 $\mu\text{g}/\text{m}^3$

Transported

~9,148 deaths/yr · ₹45,742 cr/yr

-- maintain monitoring

Compliance: 5 recommendations — all awaiting officer review

+24h 34 $\mu\text{g}/\text{m}^3$

transported-signature

stable

Ten cities on one scoreboard: PM2.5 now vs +24 h, Swachh-Vayu-style ranking, trend, dominant source, playbook recommendation, health-burden line and enforcement-compliance counts — powered by the Multi-City agent. **Click a city** and the whole console follows:

COMPARE · MUMBAI

Cities

Ten cities on one scoreboard: who is worse, what is driving it, and which playbook worked elsewhere.

- 1 Scoreboard 2 What worked elsewhere

1 Multi-City Compare

Kolkata: construction dust · Bengaluru: industrial · Mumbai: construction dust · Delhi: transported · Pune: construction dust · Lucknow: industrial · Chennai: construction dust · Ahmedabad: transported · Hyderabad: industrial · Jaipur: transported · city-average PM2.5

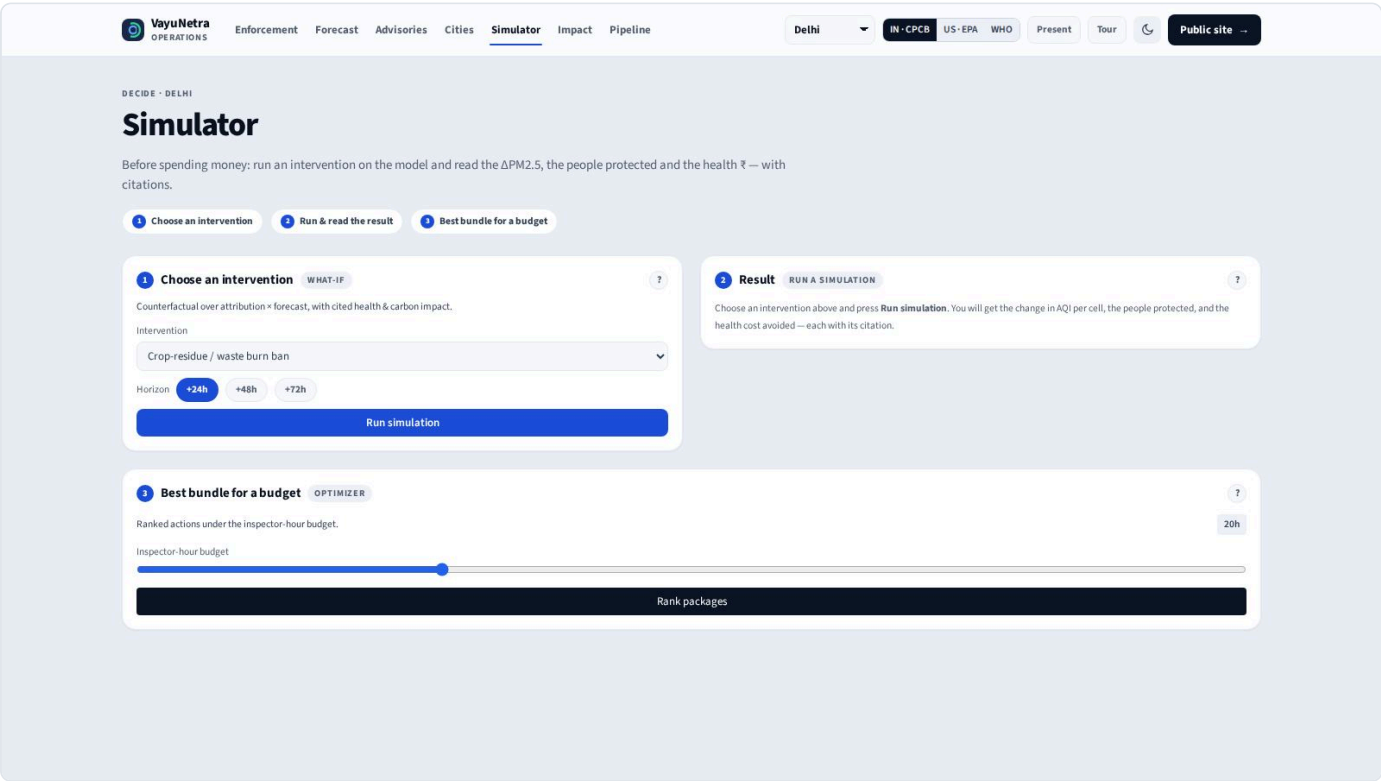
+24h

avg
now

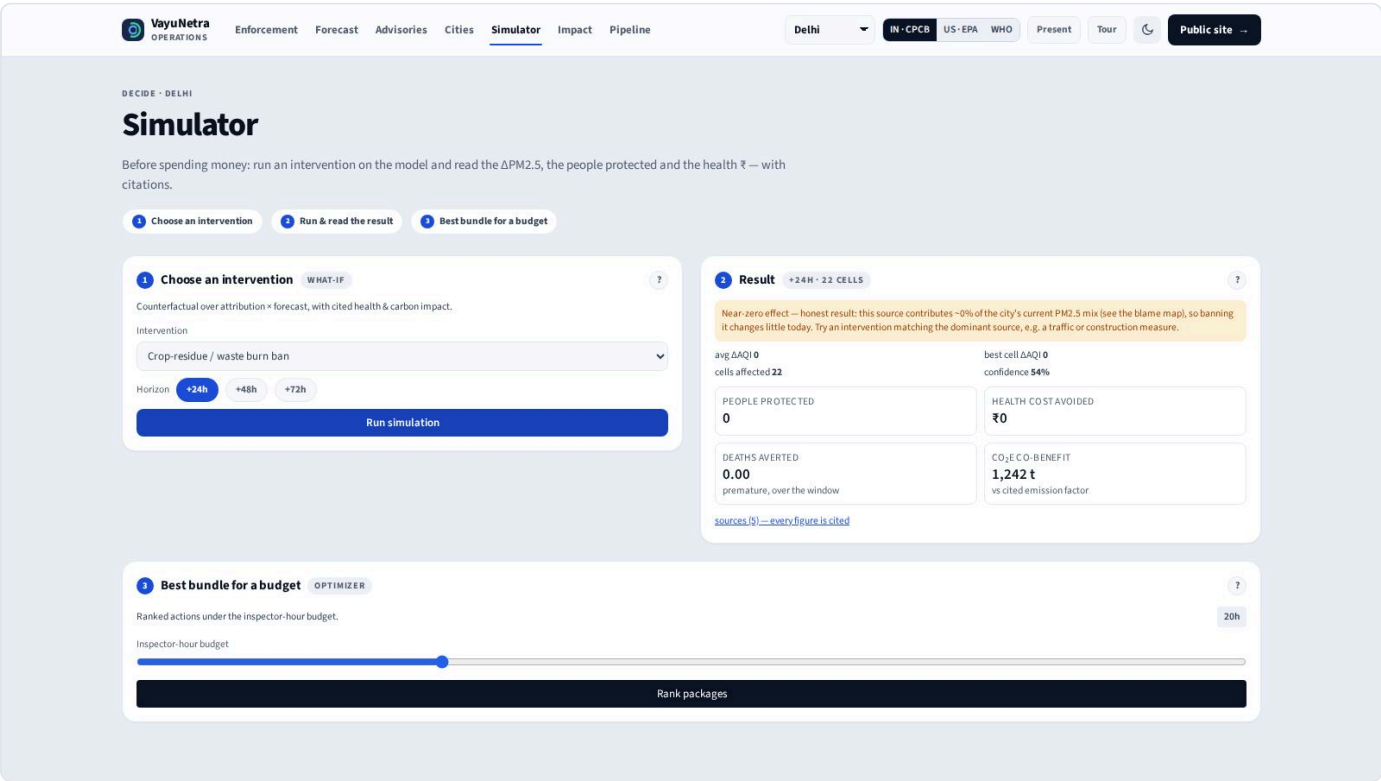
CLEAN- AIR RAN KIN G — Swachh Vayu style, by current PM2.5 (cleanest first)

#1 Mumbai 27 avg 16 $\mu\text{g}/\text{m}^3$ Construction Dust +24h 12 $\mu\text{g}/\text{m}^3$ construction-winter +40,168 deaths/yr · ₹2.01 lakh cr/yr → pre-wet exposed soil Compliance: 172 recommendations — all awaiting officer review	stable
#2 Pune 22 avg 19 $\mu\text{g}/\text{m}^3$ Construction Dust +24h 16 $\mu\text{g}/\text{m}^3$ construction-winter +12,693 deaths/yr · ₹63,466 cr/yr → pre-wet exposed soil Compliance: 14 recommendations — all awaiting officer review	stable
#3 Bengaluru 22 avg 19 $\mu\text{g}/\text{m}^3$ Industrial +24h 11 $\mu\text{g}/\text{m}^3$ industrial-signature +19,866 deaths/yr · ₹99,332 cr/yr → verify consent-to-operate limits Compliance: 120 recommendations — all awaiting officer review	stable
#4 Chennai 23 avg 20 $\mu\text{g}/\text{m}^3$ Construction Dust +24h 21 $\mu\text{g}/\text{m}^3$ construction-winter +15,838 deaths/yr · ₹79,191 cr/yr → pre-wet exposed soil Compliance: 11 recommendations — all awaiting officer review	stable
#5 Lucknow 26 avg 22 $\mu\text{g}/\text{m}^3$ Industrial +24h 19 $\mu\text{g}/\text{m}^3$ industrial-signature +10,856 deaths/yr · ₹54,278 cr/yr → verify consent-to-operate limits Compliance: 6 recommendations — all awaiting officer review	stable
#6 Hyderabad 26 avg 22 $\mu\text{g}/\text{m}^3$ Industrial +24h 19 $\mu\text{g}/\text{m}^3$ industrial-signature +18,663 deaths/yr · ₹93,316 cr/yr → verify consent-to-operate limits Compliance: 15 recommendations — all awaiting officer review	stable
#7 Kolkata 22 avg 25 $\mu\text{g}/\text{m}^3$ Construction Dust +24h 23 $\mu\text{g}/\text{m}^3$ construction-winter +34,320 deaths/yr · ₹1.72 lakh cr/yr → pre-wet exposed soil Compliance: 14 recommendations — all awaiting officer review	stable
#8 Delhi 26 avg 34 $\mu\text{g}/\text{m}^3$ Transported +24h 32 $\mu\text{g}/\text{m}^3$ transported-signature +73,395 deaths/yr · ₹3.67 lakh cr/yr highest burden → maintain monitoring Compliance: 45 recommendations - 1 dismissed	stable
#9 Ahmedabad 28 avg 35 $\mu\text{g}/\text{m}^3$ Transported +24h 26 $\mu\text{g}/\text{m}^3$ transported-signature +19,013 deaths/yr · ₹95,067 cr/yr → maintain monitoring Compliance: 11 recommendations — all awaiting officer review	stable
#10 Jaipur 25 avg 39 $\mu\text{g}/\text{m}^3$ Transported +24h 34 $\mu\text{g}/\text{m}^3$ transported-signature +9,148 deaths/yr · ₹45,742 cr/yr → maintain monitoring Compliance: 5 recommendations — all awaiting officer review	stable

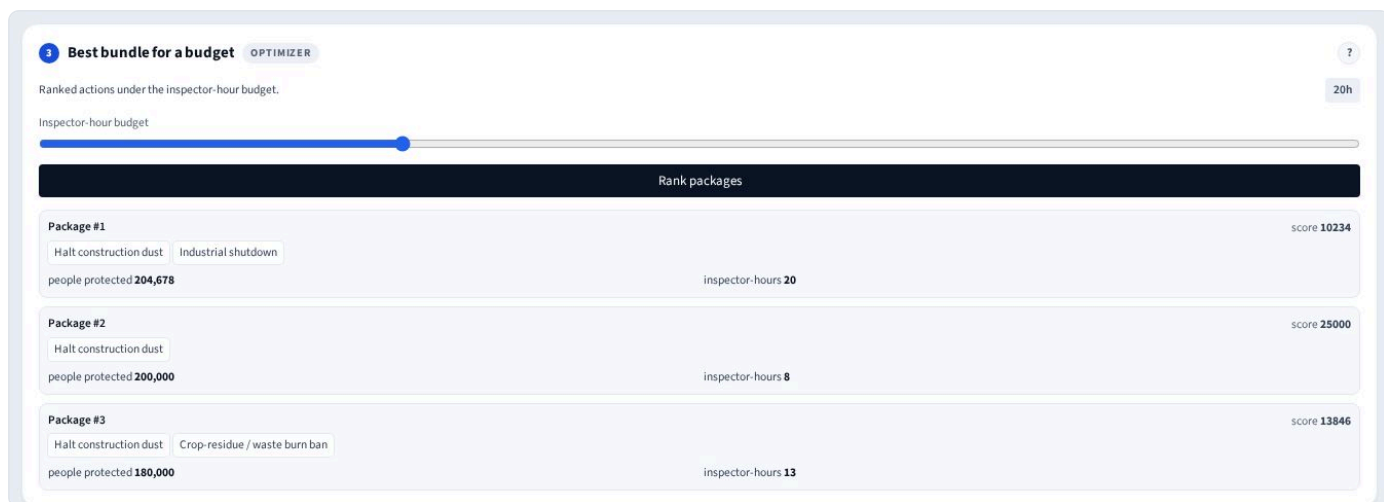
10. Simulator (section 5)



Steps: **1 Choose an intervention** · **2 Run & read the result** · **3 Best bundle for a budget**. Pick an intervention (waste-burn ban · halt construction dust · odd-even traffic · industrial shutdown · GRAP Stage III package) and a horizon → **Run simulation**.

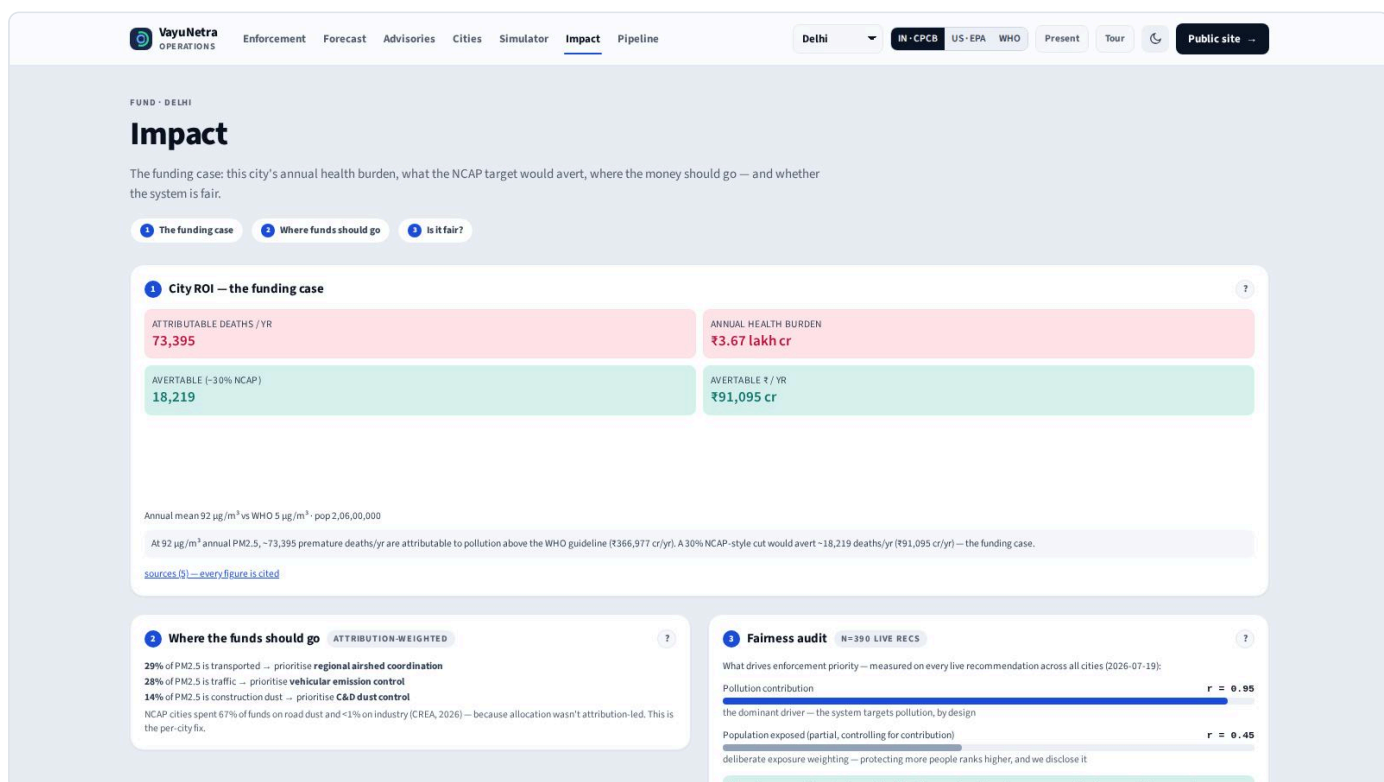


The result: ΔAQI (average and best cell), cells affected, confidence, and the cited impact cards — people protected (real population), health cost avoided (₹), deaths averted, CO₂e co-benefit — every figure with its citation. **Honest zero**: if the chosen source barely contributes today the app says so instead of inventing an impact.



Best bundle for a budget: set an inspector-hour budget → **Rank packages** → intervention bundles ranked by impact per inspector-hour.

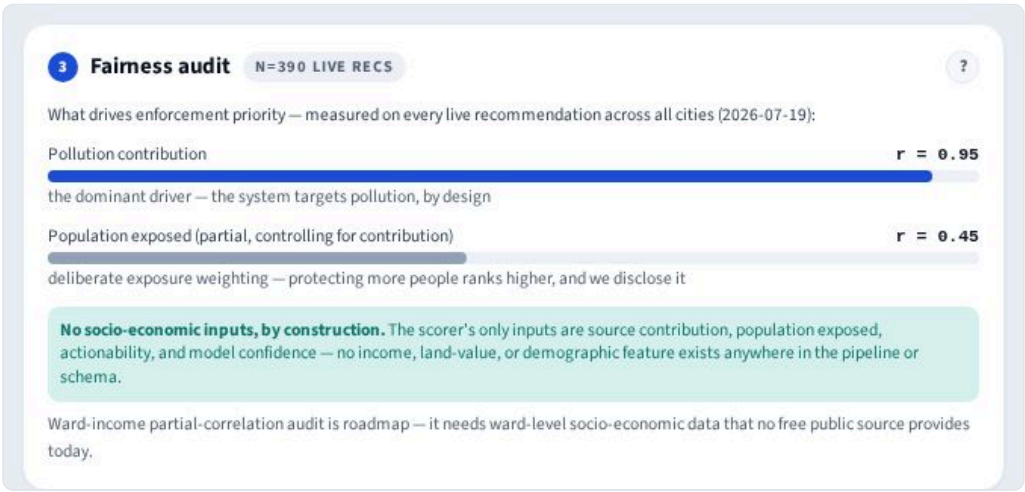
11. Impact (section 6)



Steps: **1 City ROI** • **2 Where the funds should go** • **3 Fairness audit**. - **City ROI** — attributable deaths / yr, annual health burden (₹), what a 30 % NCAP cut would avert, with citations and the VSL caveat.

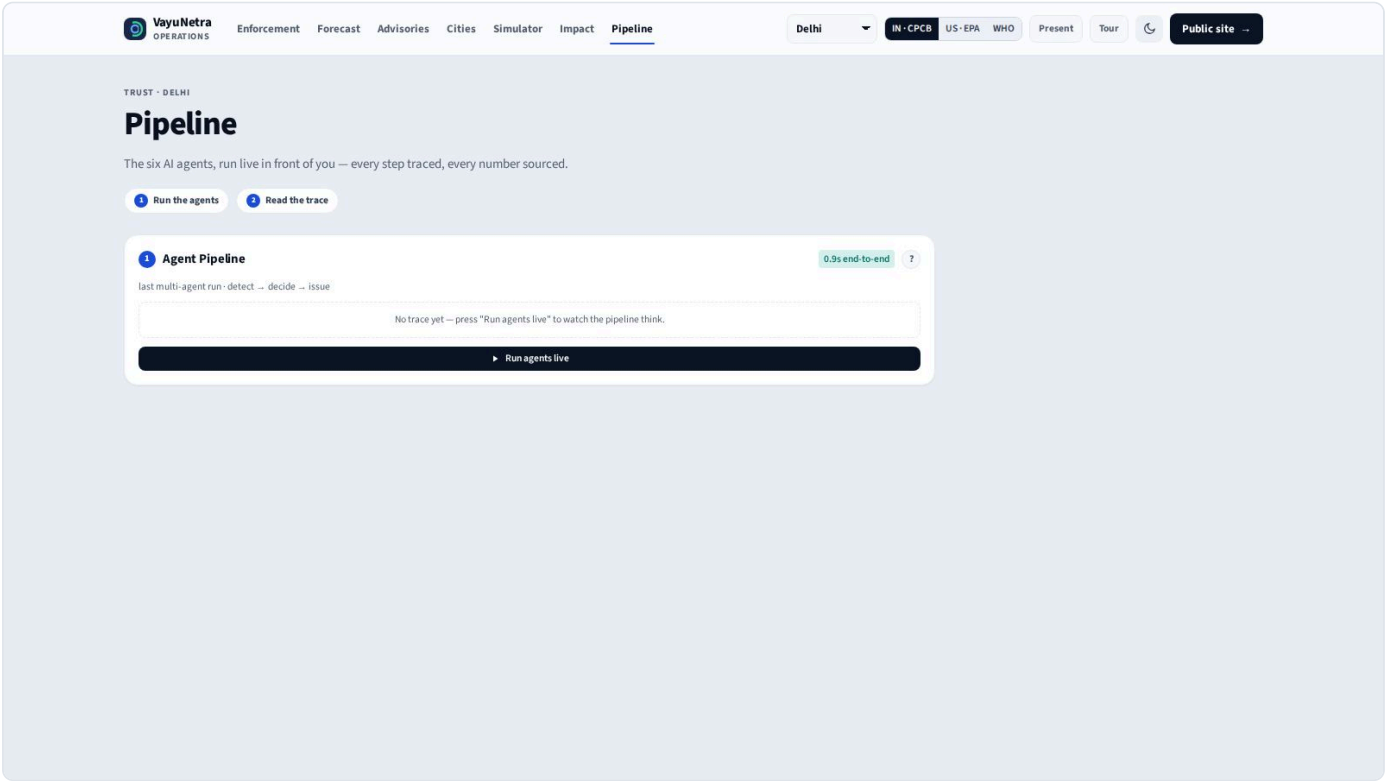


- **Where the funds should go** — live source shares mapped to NCAP spending heads (attribution-led allocation, the answer to CREA's finding that cities put 67 % of funds into road dust).

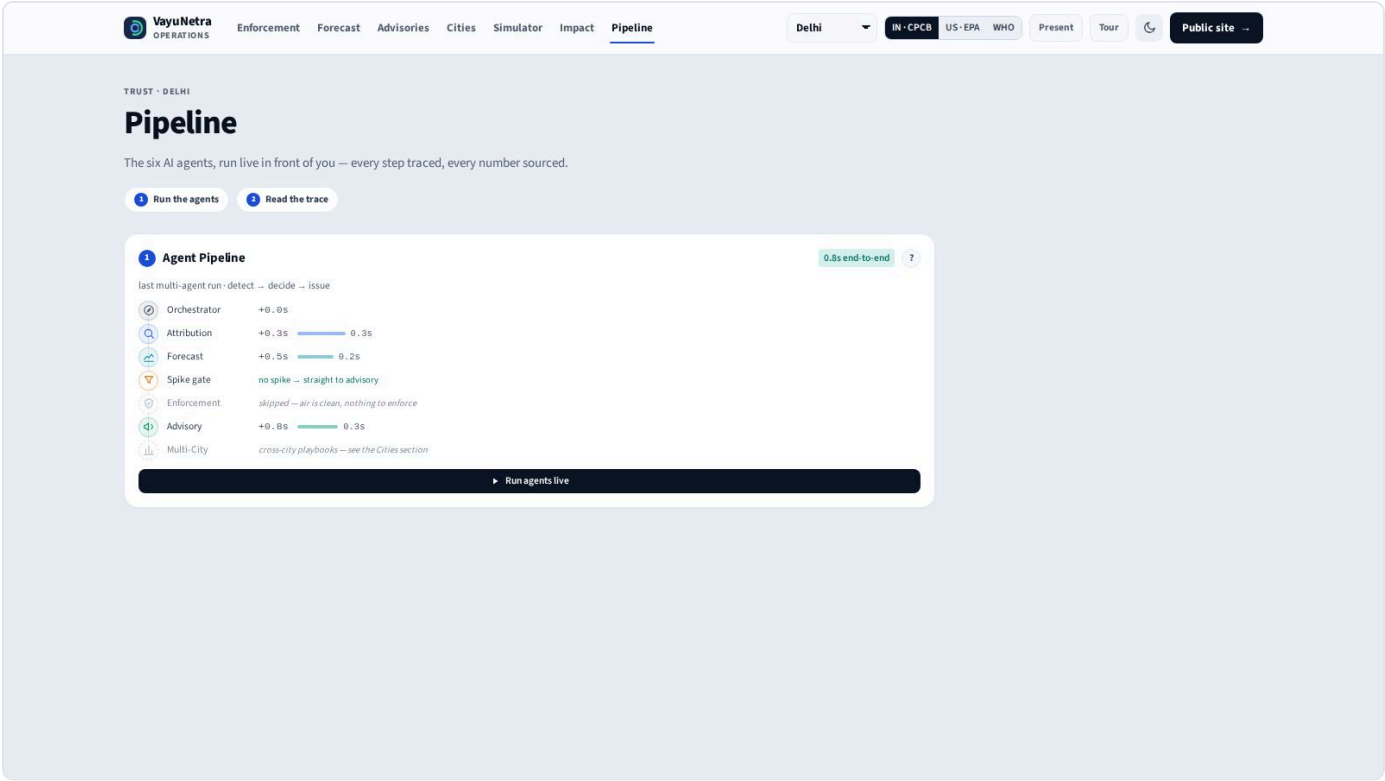


- **Fairness audit** — enforcement priority correlates with pollution contribution (by design) and people exposed (disclosed); **no socio-economic input exists anywhere in the pipeline.**

12. Pipeline (section 7)



Steps: **1 Run the agents** · **2 Read the trace**. The last multi-agent run as a timeline — Orchestrator → Attribution → Forecast → **Spike gate** → Enforcement → Advisory — with per-node durations. The spike gate is a decision: no spike → straight to advisory; spike → escalated to enforcement (on clean air Enforcement shows *skipped* rather than pretending). **Run agents live** triggers a real end-to-end run against the live database and updates the trace in front of you (0.8–9.7 s of compute).



13. Under the hood

13.1 The six agents (LangGraph)

Agent	Does
A0 Orchestrator	Reads latest signals, finds spiking cells, routes the graph, stamps per-node latency traces
A1 Attribution	GBM + SHAP hybrid source attribution per cell; abstains to signature priors below the R^2 gate
A2 Forecast	24/48/72 h per-cell PM2.5, persistence-blended, CQR-calibrated intervals, calibrated exceedance probabilities
Spike gate	Branches: enforcement only when cells spike ($PM_{2.5} > 120 \mu g/m^3$ or focus cells exist)
A3 Enforcement	Priority = contribution $\times$ exposure $\times$ actionability $\times$ confidence; jurisdiction-aware RAG citations; writes the worklist without touching acted-upon items
A4 Advisory	Deterministic multilingual advisories, vulnerability-adjusted per ward/audience
A5 Multi-City	Cross-city trends, dominant sources, playbooks, health burden

Supporting models: E1 satellite source detection (labelled Earth-Engine heuristic; a learned detector is roadmap) · E2 dense 1-km field · E3 counterfactual what-if · E5 optimiser · E7 health ₹ / lives / CO₂e · GRAP forecast trigger · multimodal RAG (text + Sentinel-2 patches).

13.2 Data sources (all free / open)

CPCB CAAQMS (via OpenAQ / data.gov.in) · Open-Meteo + ERA5 weather and boundary layer · Sentinel-5P NO₂, Sentinel-2, NASA FIRMS (Earth Engine) · OpenStreetMap sources / facilities / roads · GPW v4.11 population · GTFS + road-network mobility proxy. Ingest and model refresh run on GitHub Actions (hourly + nightly); one row per reading (unique key), raw readings kept 180 days then archived to Storage with a daily rollout kept forever (docs/SCALE.md).

13.3 API (FastAPI on Render)

44 HTTP routes + 1 WebSocket, one envelope. Feeds (/cities /aqi/current /history* /attribution /forecast /coverage /plume /static-layers ...), officer (/enforcement* , /interventions* , /brief*), citizen (/advisory* , /clean-zones , /telegram/webhook , /ivr/* , /report*), analysis (/comparison /roi /simulate /optimize /latency /traces), metrics (/metrics/benchmark /metrics/attribution /metrics/interventions), /landing/snapshot , agents/admin (/agent/query , /admin/cities , WS /live). Full table with roles: docs/API_CONTRACT.md . Public reads via anon JWT under RLS; every write server-side with the service role; admin behind X-Admin-Key .

13.4 Database (Supabase Postgres + PostGIS + pgvector)

cities, measurements, attribution, forecasts, emission_sources, enforcement_recgs, enforcement_status_log, intervention_tracking, advisories, advisory_subscribers, citizen_reports, kb_chunks, action_traces, coverage_field, vulnerability, profiles, pm25_daily_rollover — 20 migrations in supabase/migrations/ .

13.5 Honesty, by construction

- Attribution **abstains** (R² gate + coverage mask) instead of guessing; how today's split was made is shown in the console.
- The forecast is benchmarked the way a reviewer would (strict temporal split, production-faithful rolling refit, three baselines, shared support mask, regime slices, onset recall, Brier skill) and the numbers — including the Severe-tail weakness — are served by the API and printed in the console.
- Attribution is compared bucket by bucket with published apportionment (TERI-ARAI 2018; Guttikunda 2019, CSTEP 2022) with disagreements stated (docs/ATTRIBUTION_VALIDATION.md).
- Real GRAP orders are replayed and a weather-normalised effect check is published, null results included (docs/OUTCOMES.md).
- Health text is LLM-free; impact figures return null over invented constants; notices are drafts; nothing fabricated is written to the production database; the audit trail is append-only.
- Where ₹0 stops is measured and published (docs/SCALE.md).

13.6 Ops

make dev · make test (204 backend tests, 64 % line coverage, CI gate 55 %) · cd web && npx playwright test (7 smoke; VN_LIVE=1 adds the 9-flow live officer journey) · web/scripts/qa/*.mjs (full walkthrough, axe, mobile, city sweep, deck render) · make prewarm (GO / NO-GO) · make benchmark · GitHub Actions: hourly ingest, nightly models / registry / enforcement / advisories / brief / retention.

14. Troubleshooting & FAQ

Symptom	Meaning / fix
Amber "backend waking up" notice	Render free-tier cold start; the console shows bundled fixtures meanwhile. retry or wait — it clears on the first successful call; the notice never blocks the map controls. make prewarm before demos.
Map loads, panels empty	Backend still waking, or a city with thin data — switch city once.
"No per-cell forecast" in the Cell Story	That cell has no fresh per-cell rows; the city Forecast section still works.
Worklist says nothing to enforce	Air is clean (spike gate) or filters exclude everything.
Advisory empty in a language	That city does not publish that language (languages come from the city config).
Notice PDF takes seconds	Live dossier assembly (RAG + satellite patch) — expected.

Symptom	Meaning / fix
Tour will not reappear	Once-only by design; replay with ? .
Simulator says "near-zero effect"	Honest result — pick the dominant source.
Real or demo?	<code>/health</code> shows <code>DEMO_MODE</code> ; the landing page and console say "as of" and "snapshot" where a fixture is shown.

Glossary. **H3** — Uber's hexagonal grid; res-8 $\approx$ 1 km². **SHAP** — per-feature contribution of a prediction, in $\mu\text{g}/\text{m}^3$. **CQR** — conformalised quantile regression; makes the 80 % band cover $\sim$ 80 %. **P(>120)** — calibrated probability the cell exceeds Very Poor. **Persistence** — the "tomorrow = today" baseline. **Onset recall** — share of clean→Very-Poor transitions the alarm flags in advance. **GRAP** — Delhi-NCR's Graded Response Action Plan (Stages I–IV). **RAG** — retrieval over the regulation corpus (citations, not free text). **RLS** — Postgres row-level security. **PRANA** — the NCAP reporting portal cities report actions and spend to. **GPW** — Gridded Population of the World v4.11.

15. Repo map

Path	What
<code>api/main.py</code>	All FastAPI routes
<code>agents/</code>	The six LangGraph agents, the brief, the notice PDF writer
<code>ml/</code> , <code>core/</code>	Models, benchmark and hindsight harnesses, H3 utils, impact and intervention math, city configs, ward lookup
<code>connectors/</code>	CPCB / OpenAQ, Earth Engine, Open-Meteo, OSM, population, vulnerability, mobility
<code>scripts/</code>	prewarm, refresh_advisories, archive_measurements, morning brief, fetch_history, doc PDFs
<code>web/src/</code>	React console + landing (<code>App.tsx</code> shell, one panel per card, <code>console/flows.tsx</code> = the spines)
<code>web/scripts/qa/</code>	Playwright QA: full walkthrough (this guide's screenshots), axe, mobile, city sweep, deck render
<code>demo/fixtures/</code>	The 19 JSON fixtures behind <code>DEMO_MODE</code>
<code>supabase/migrations/</code>	Schema, RLS, seeds, closure log, one-row-per-reading key, retention rollup (20 migrations)
<code>docs/</code>	This guide (<code>guide/</code> = its screenshots), the finale deck (<code>VayuNetra_Pitch.html</code> + static exports, <code>PITCH_SCRIPT.md</code> , built from <code>docs/pitch/</code>), benchmarks and outcomes, API contract, validation, scale, QA evidence
<code>eval/evaluate.ipynb</code>	The validation trail, including the documented failures